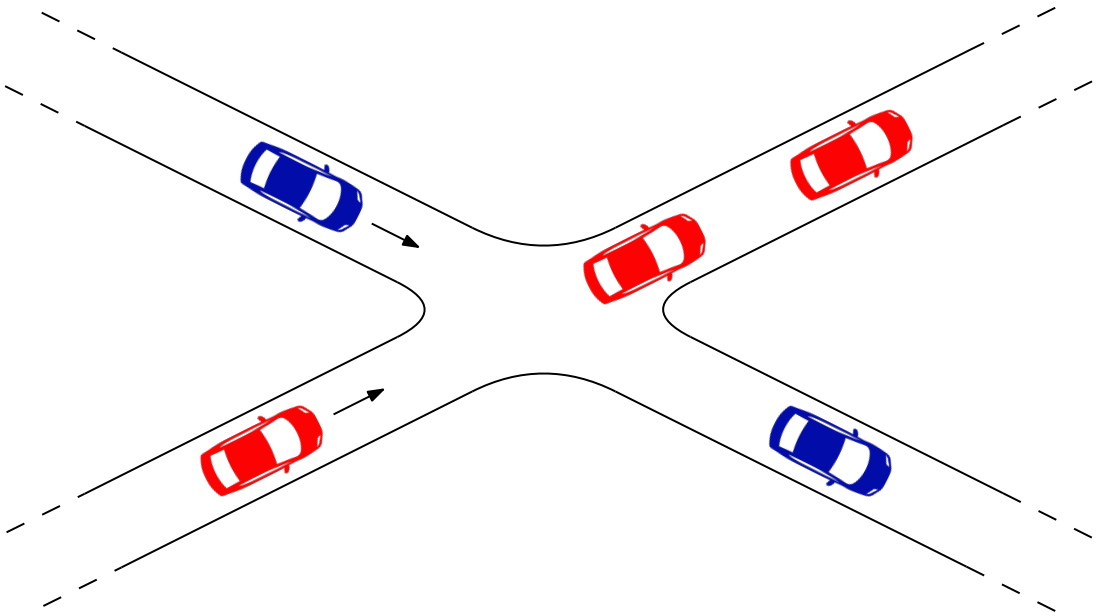
Efficient and Provably Safe Autonomous Intersection Coordination

Combinatorial Learning for Vehicle Motion Planning

Jeroen van Riel

November 10, 2025

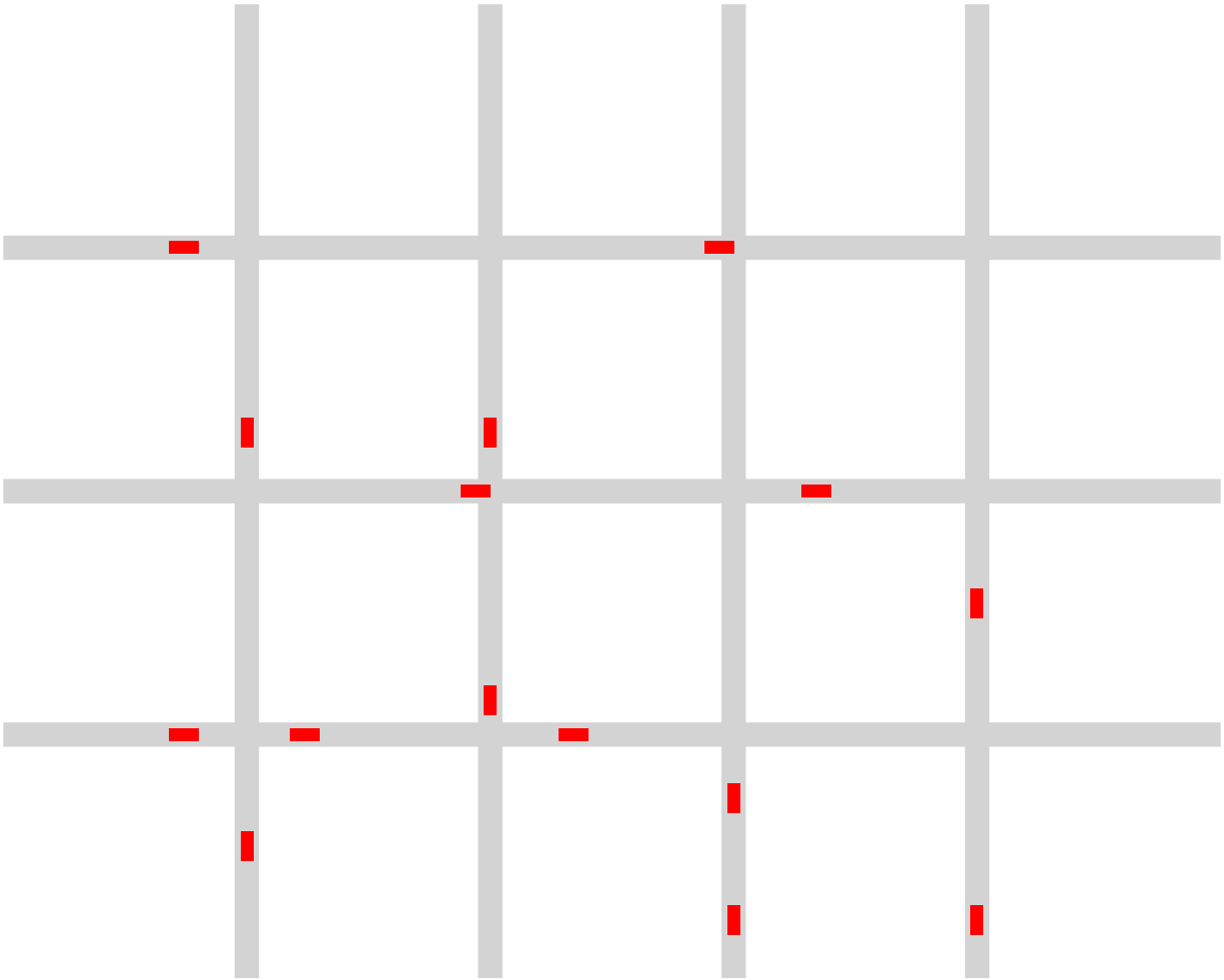
Motivation



+ safety
+ comfort

− travel time
− fuel usage

Network Model



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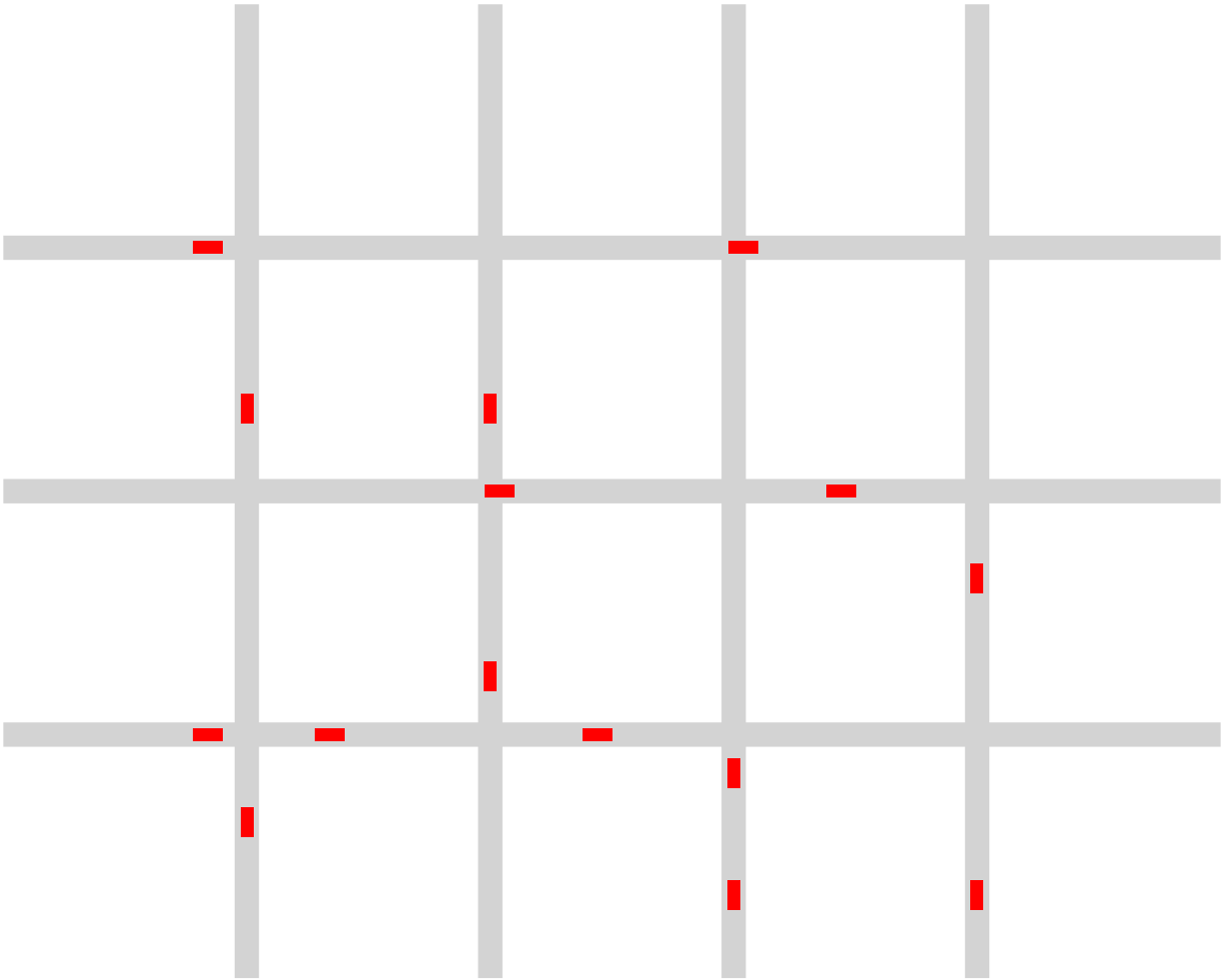

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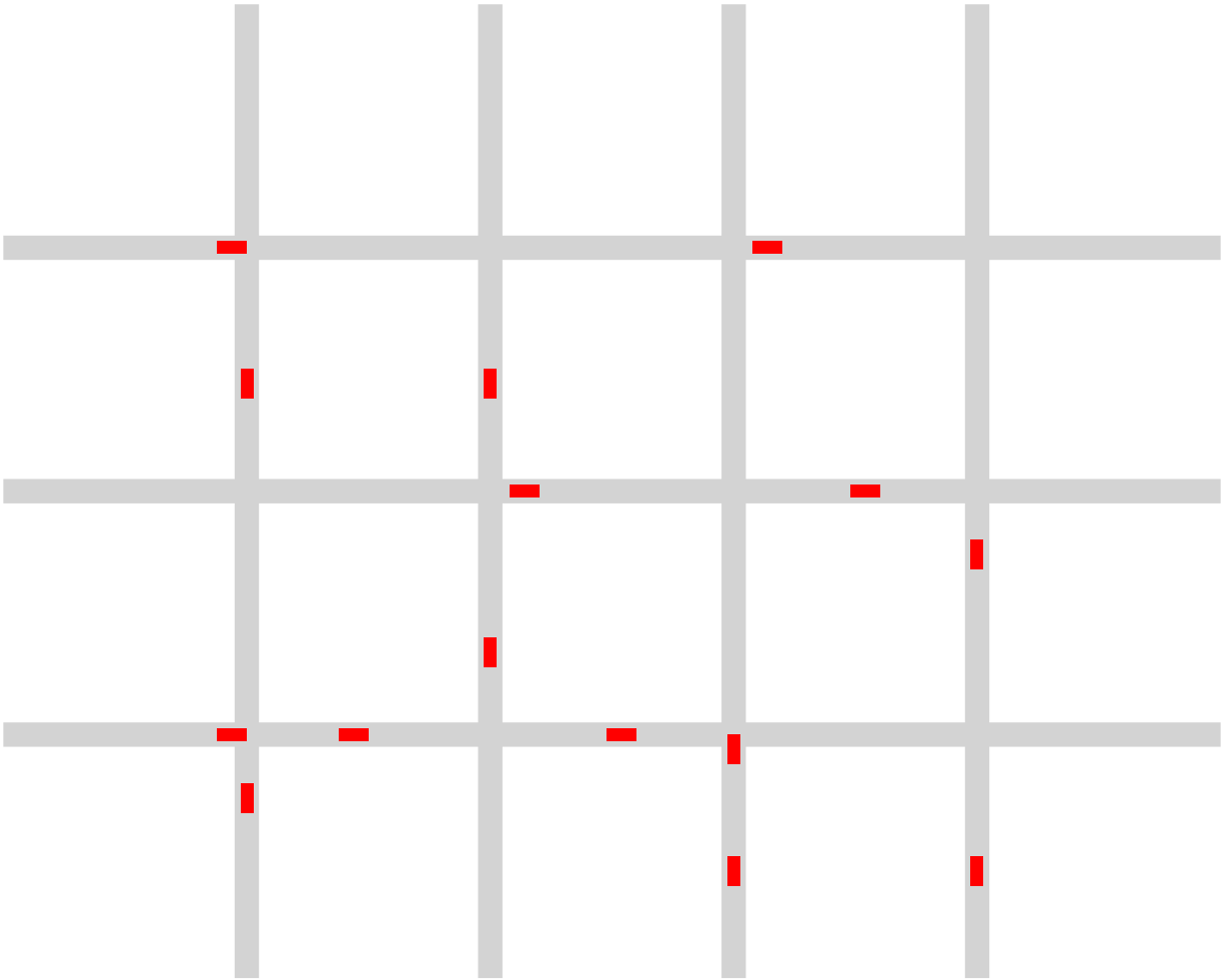

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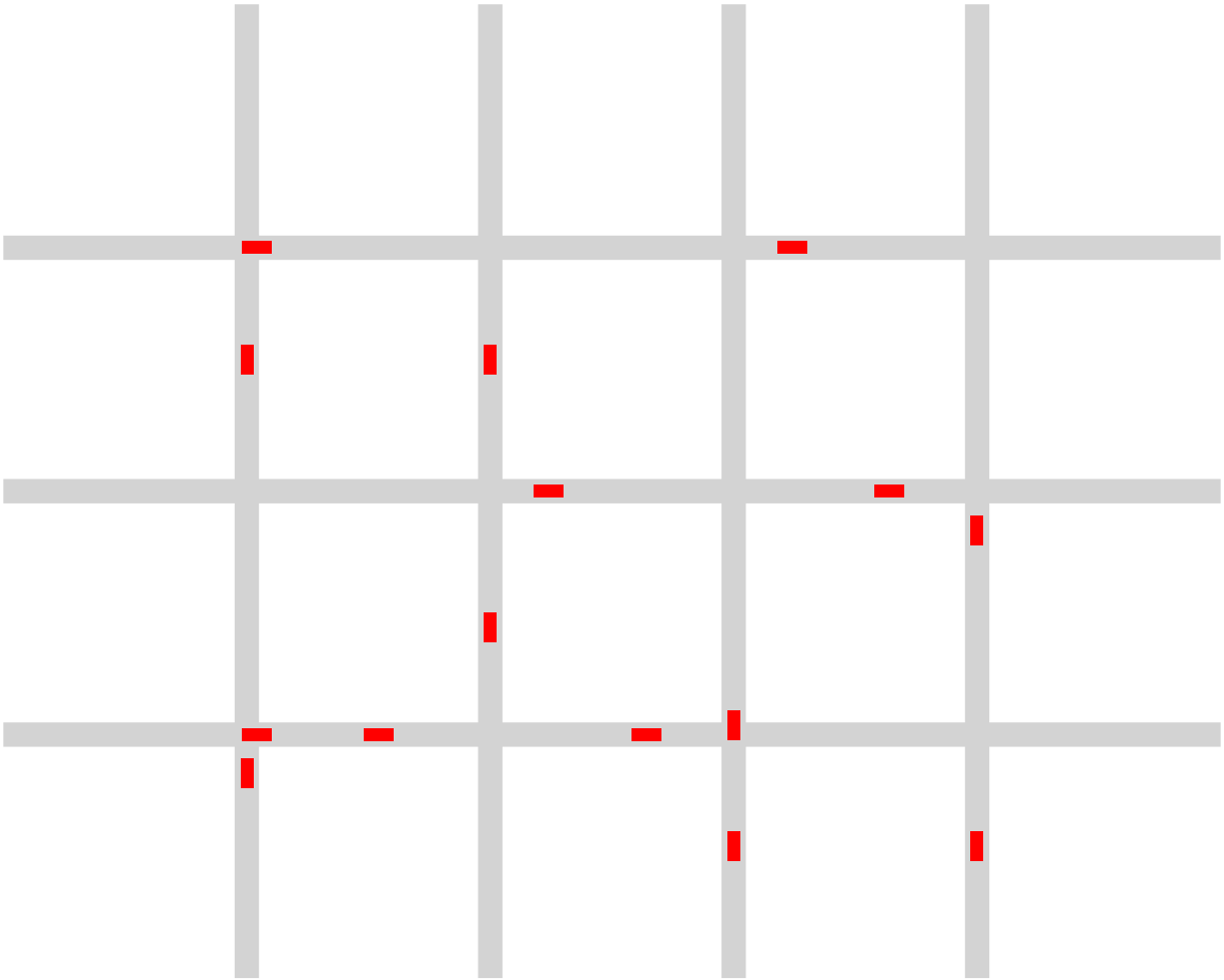
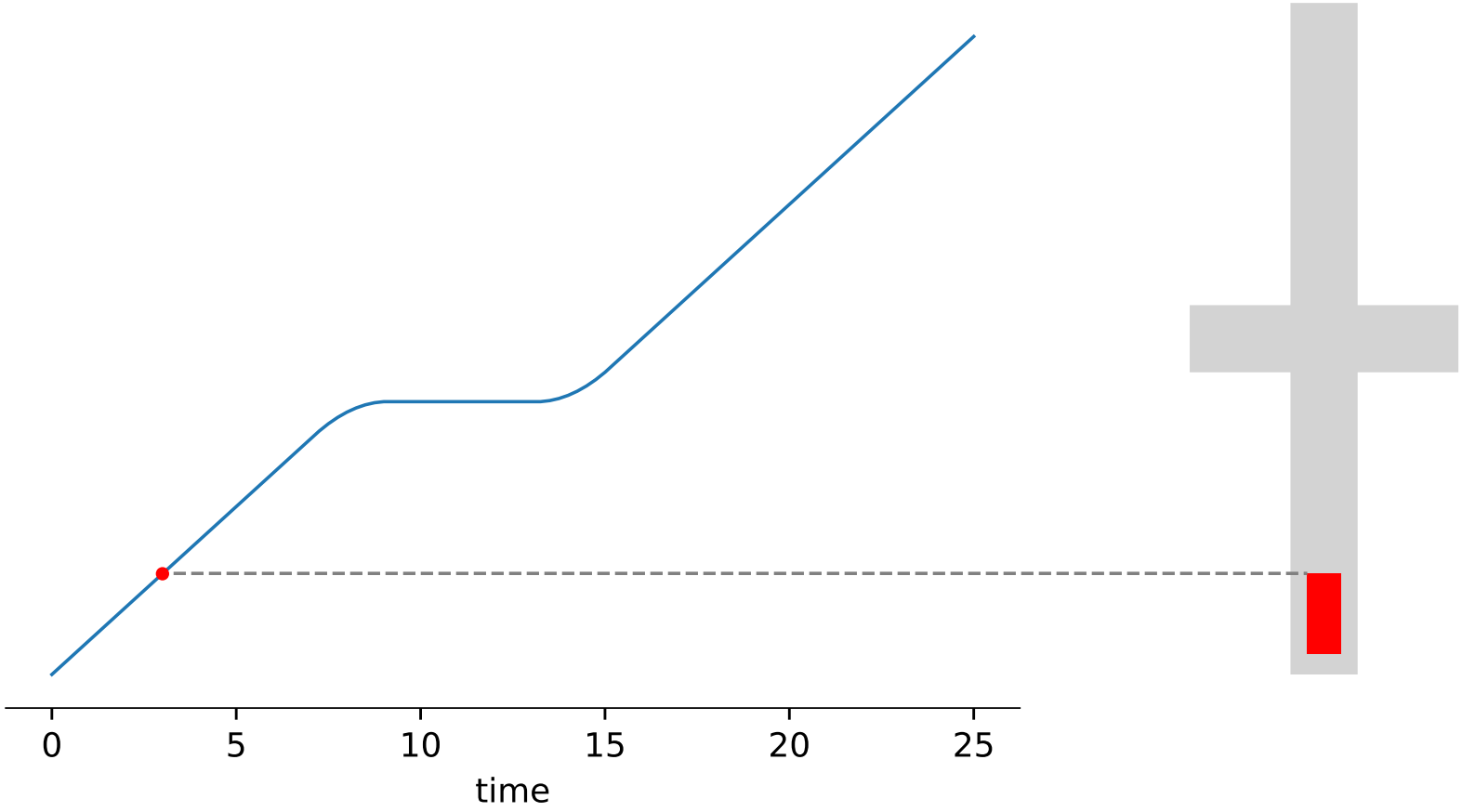


Outline

1. Crossing Time Scheduling
2. Cutting Planes in Single Intersection
3. Heuristics and Combinatorial Learning

Crossing Time Scheduling

Crossing Time



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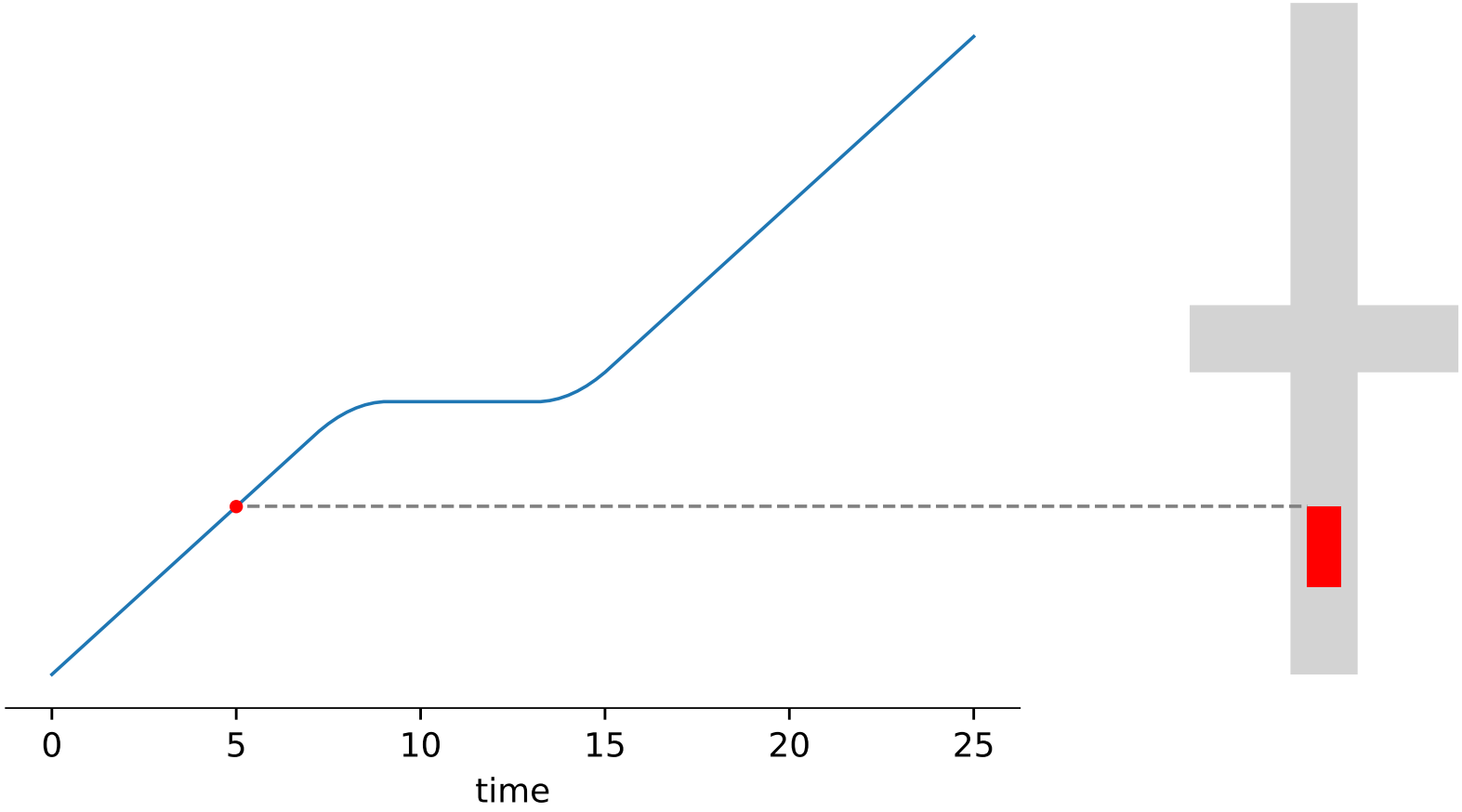

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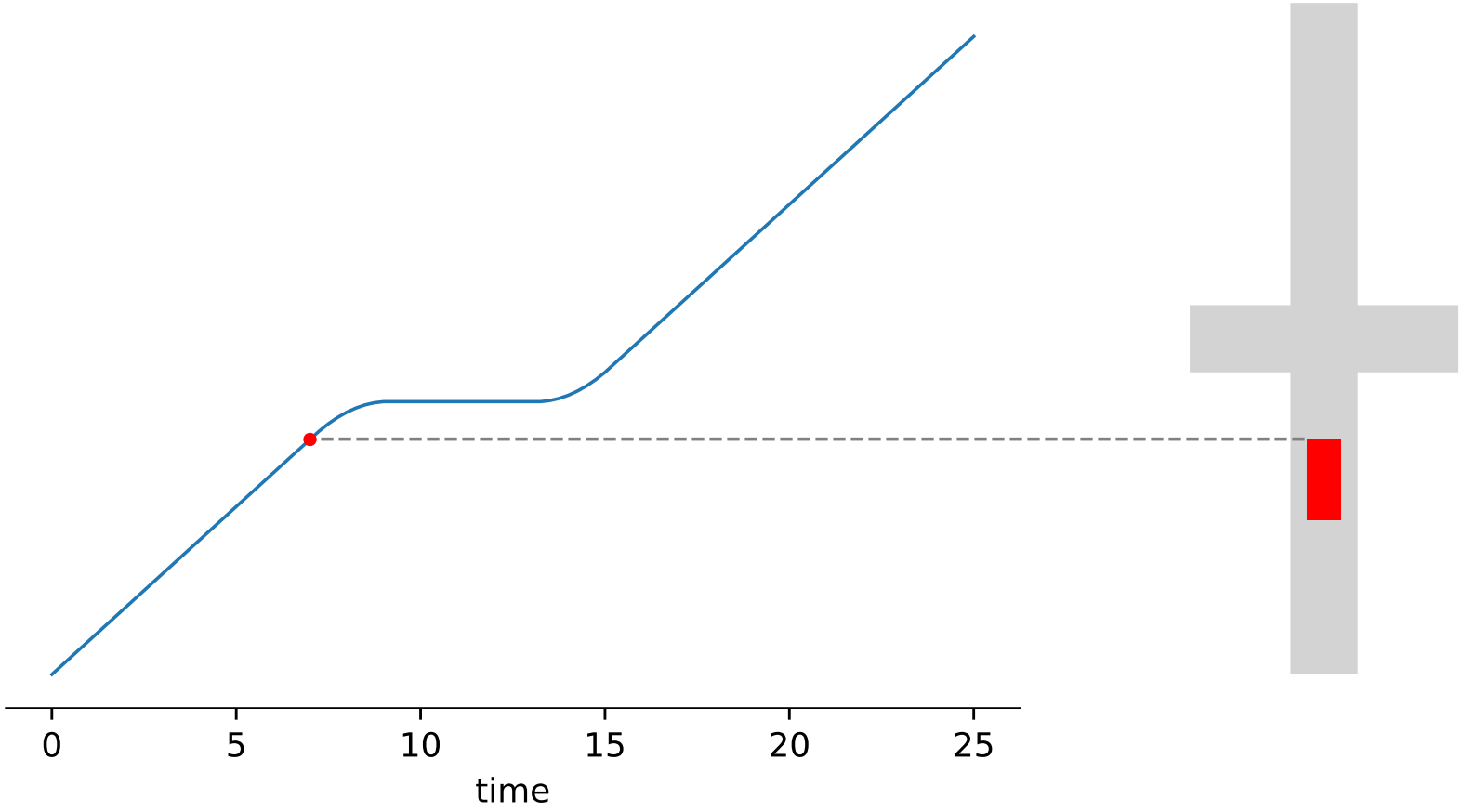

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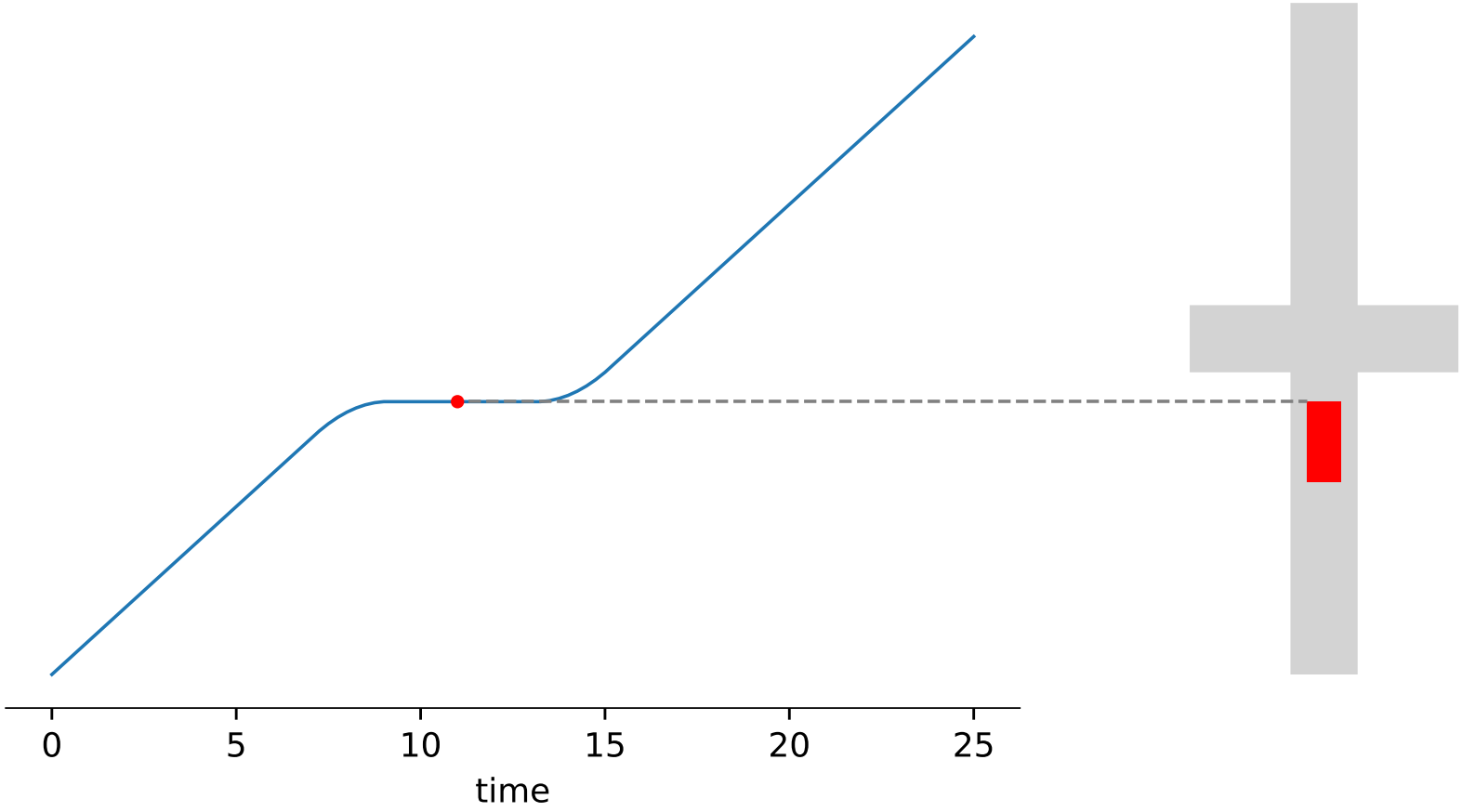

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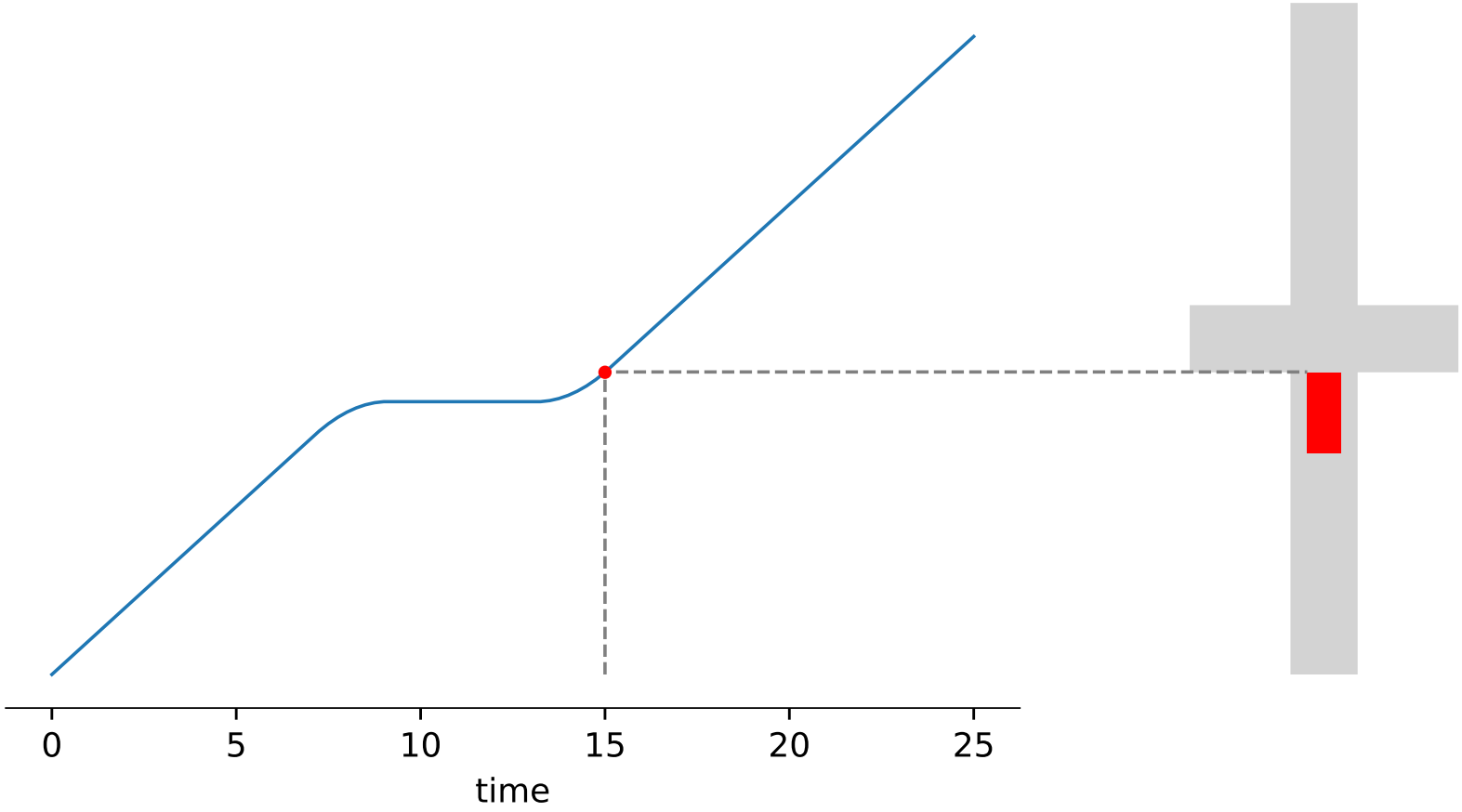

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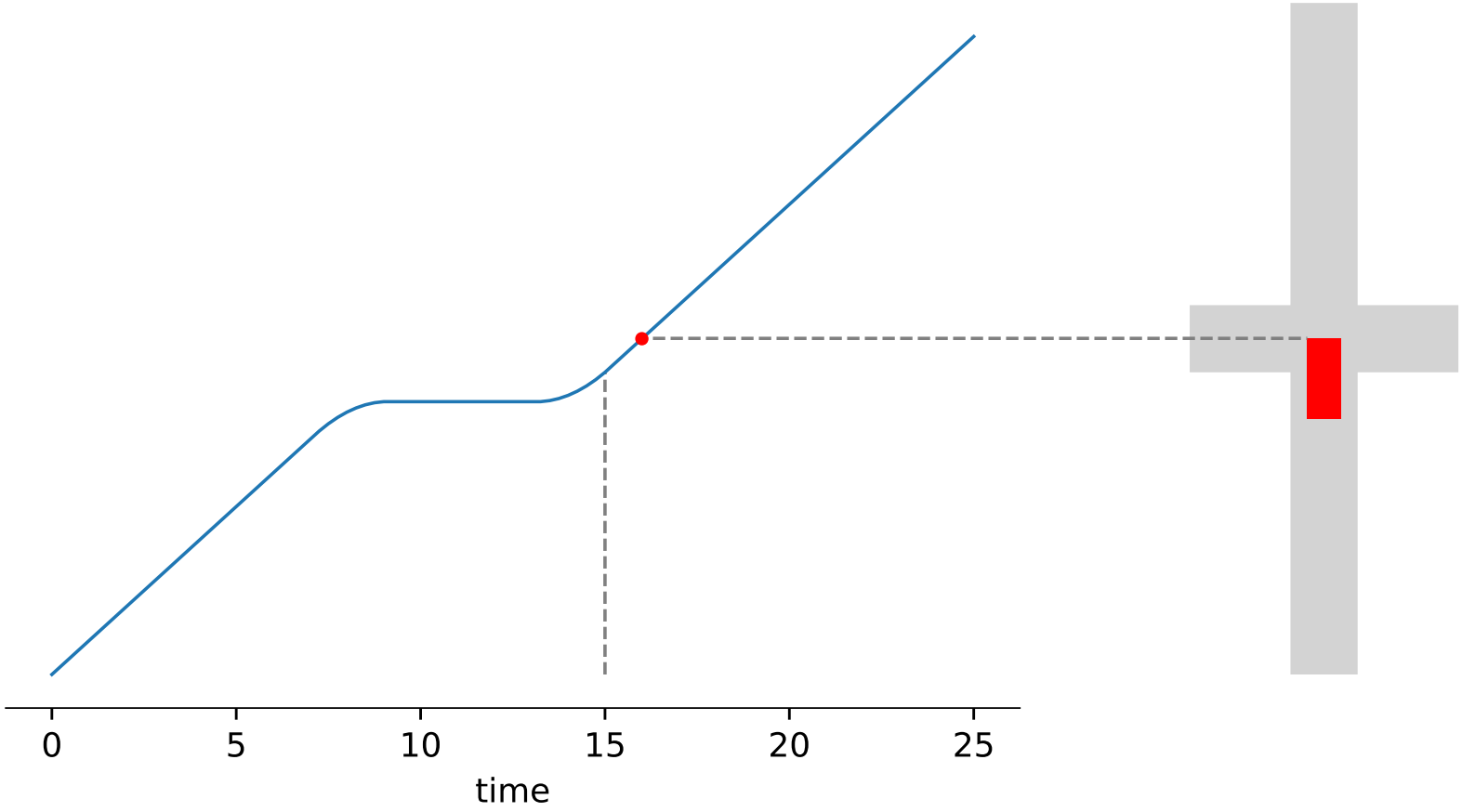

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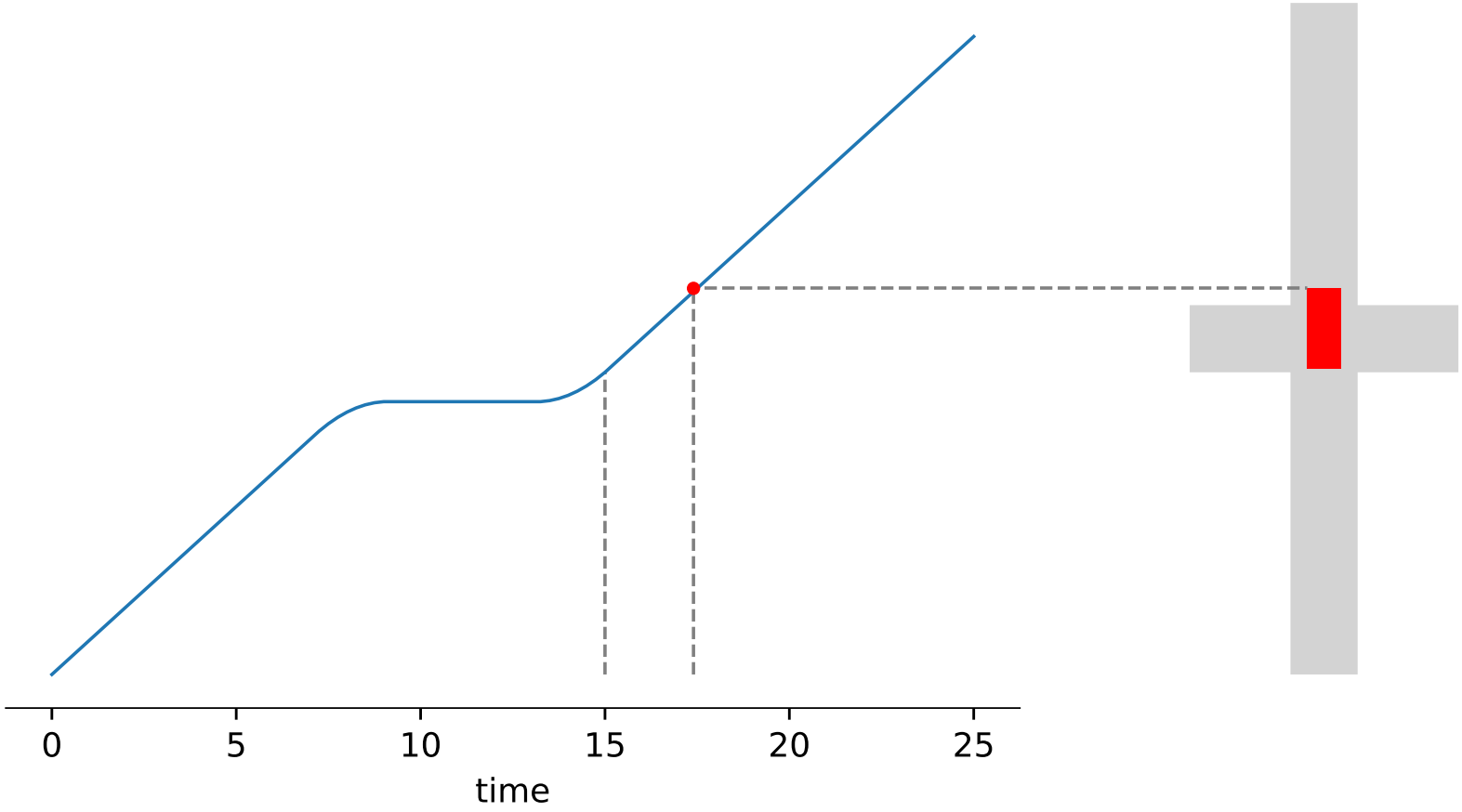

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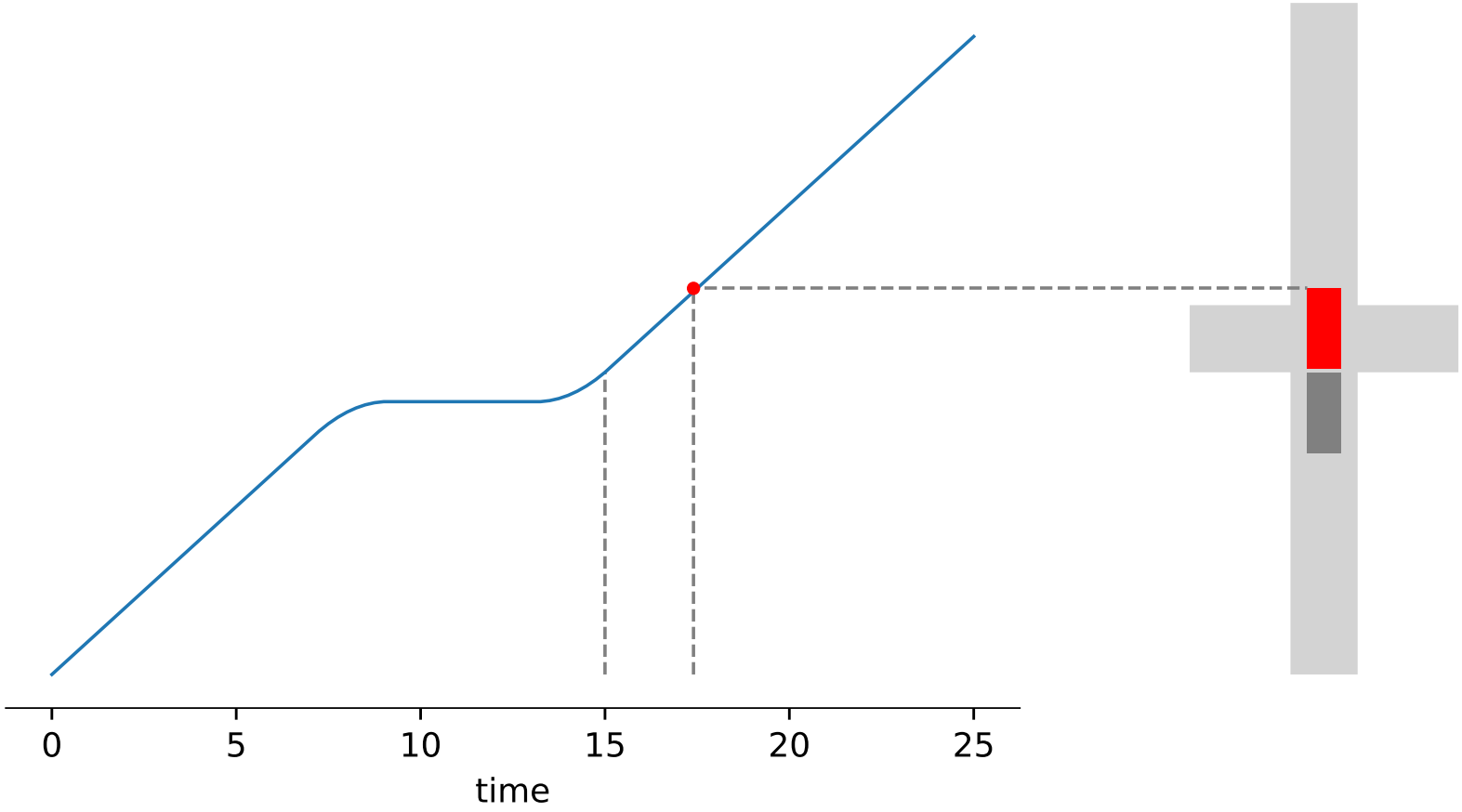

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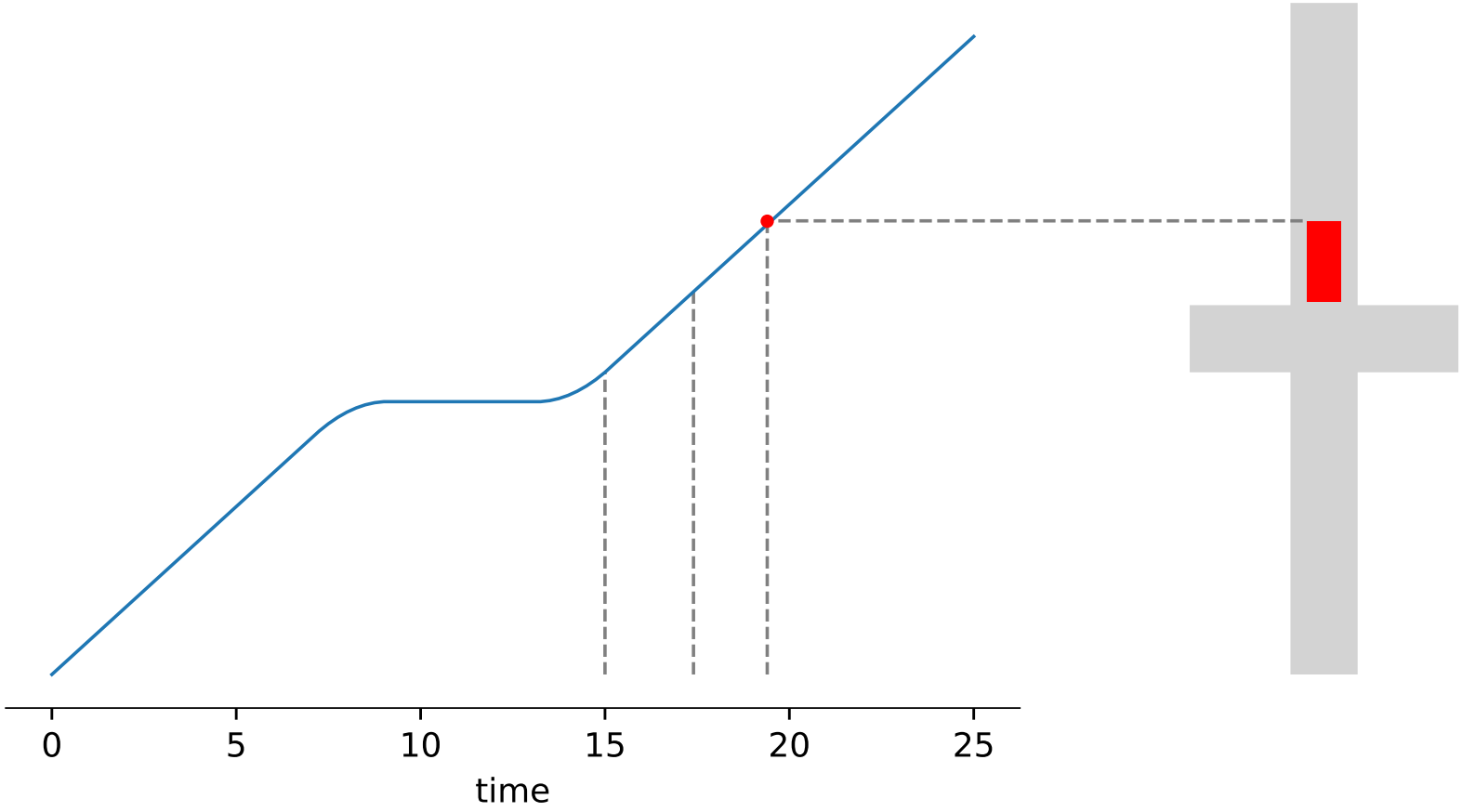

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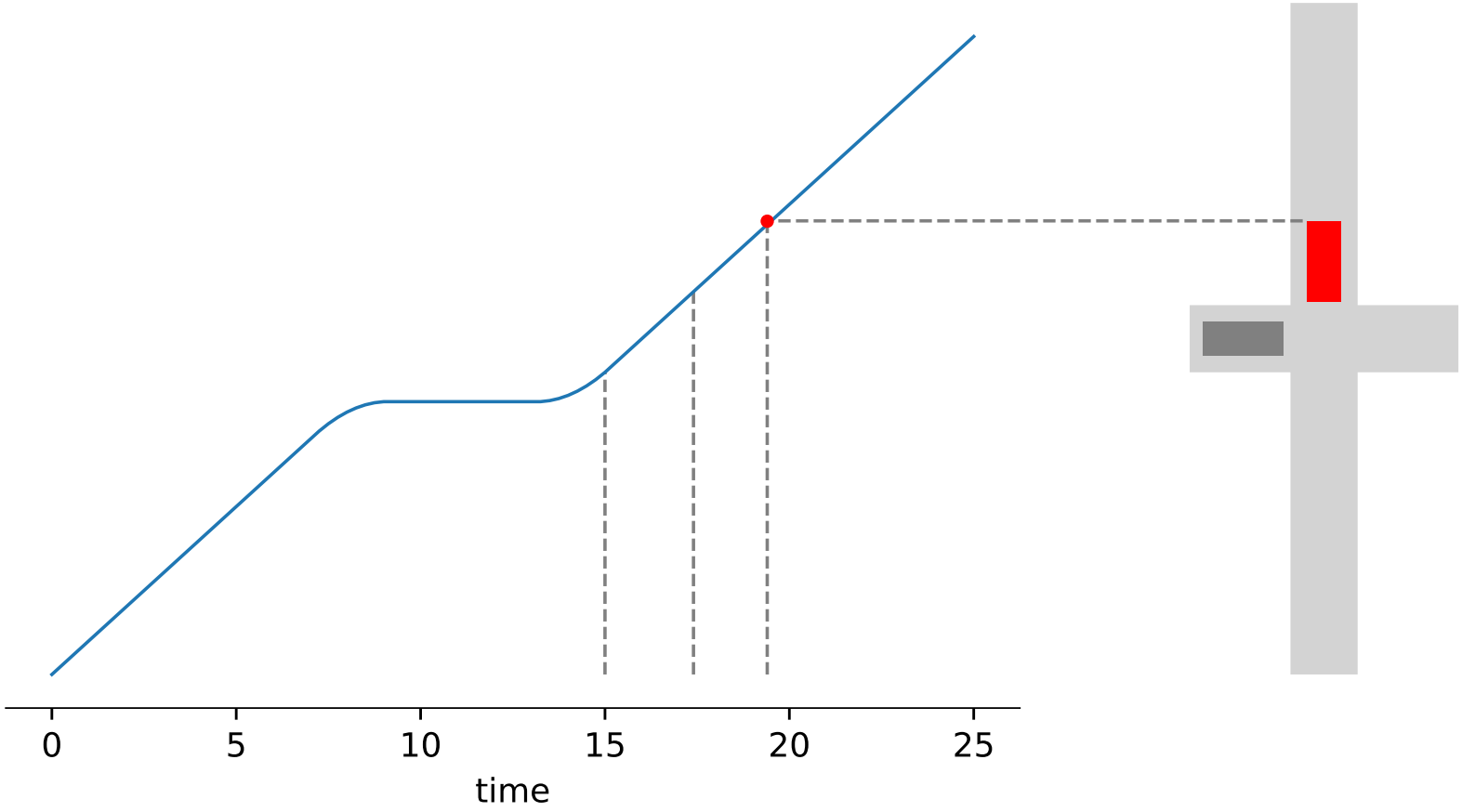

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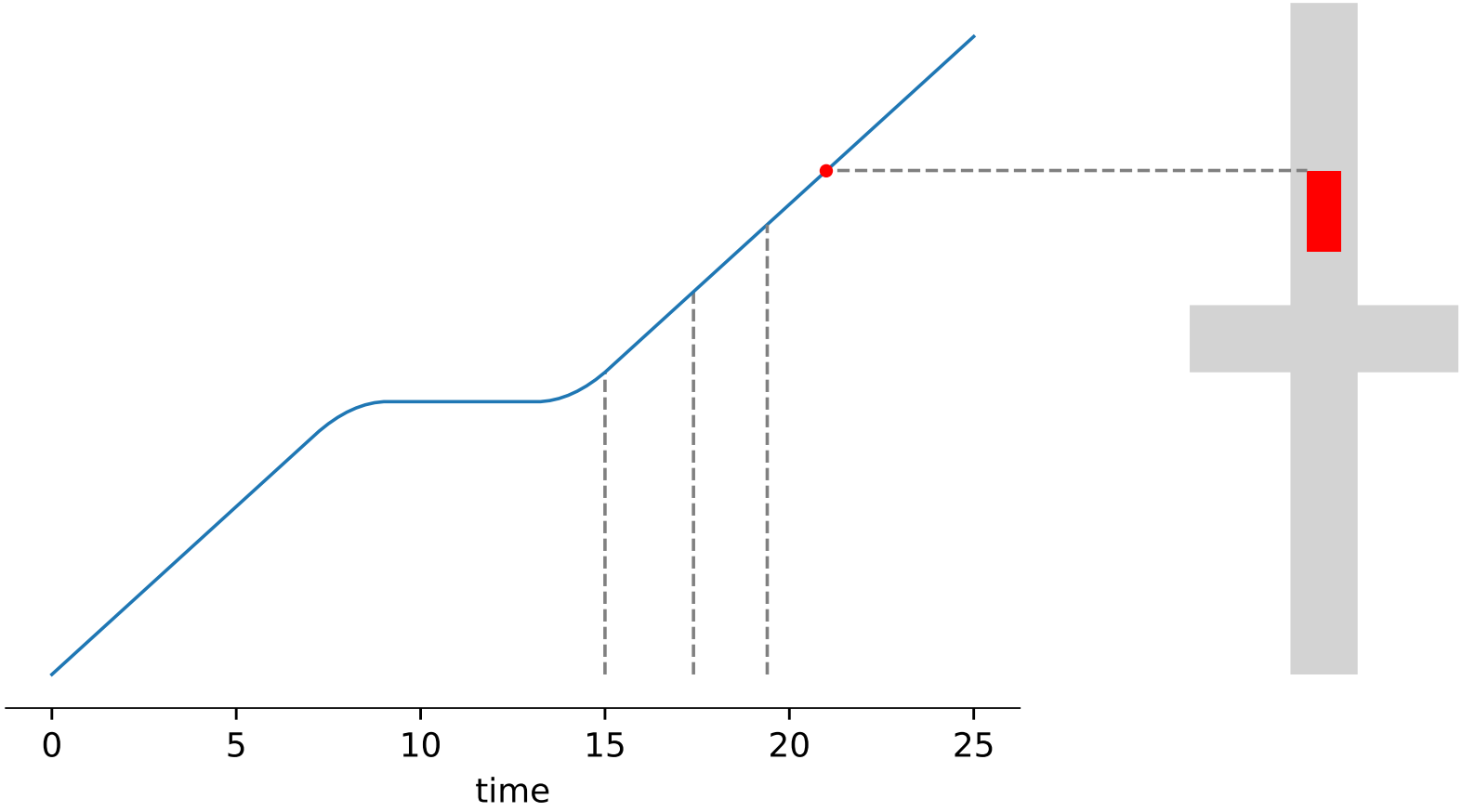

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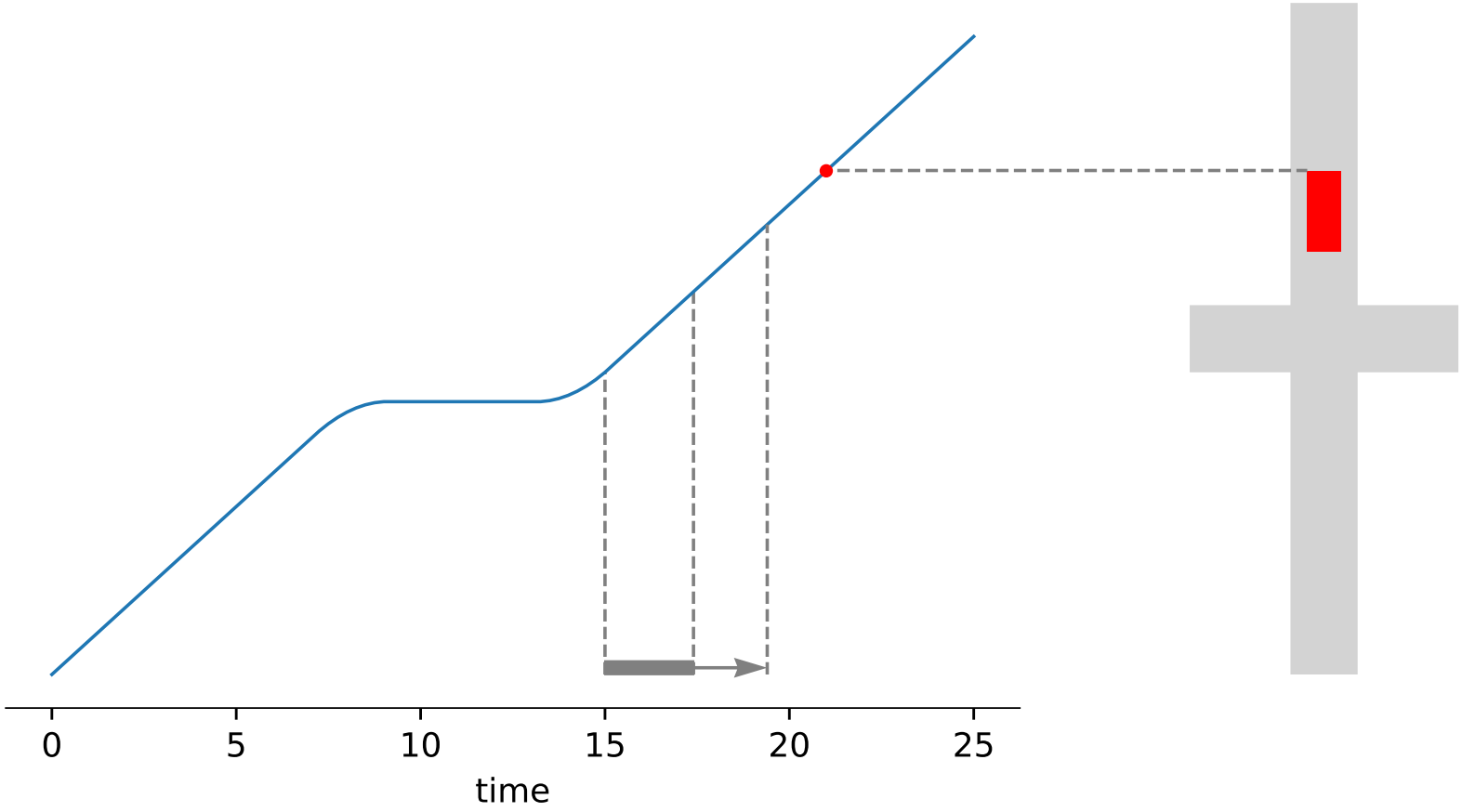
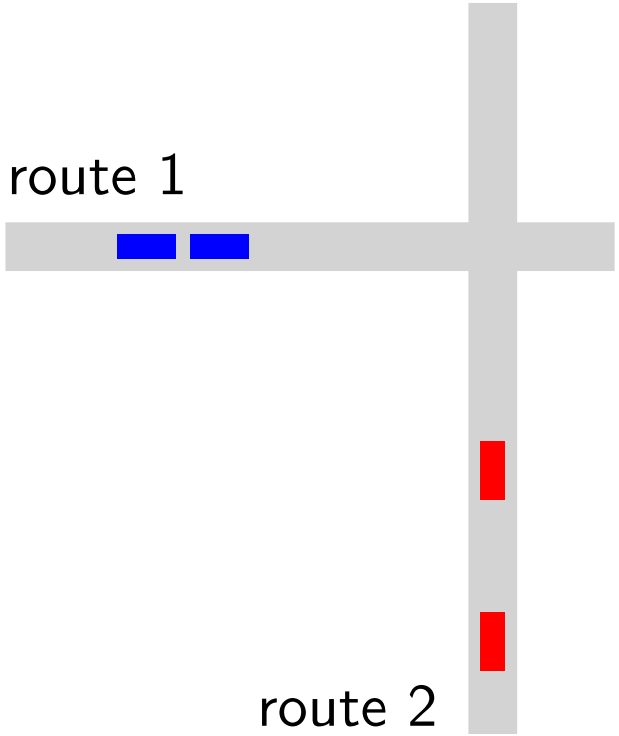


Crossing Time Scheduling

- Single intersection with two routes



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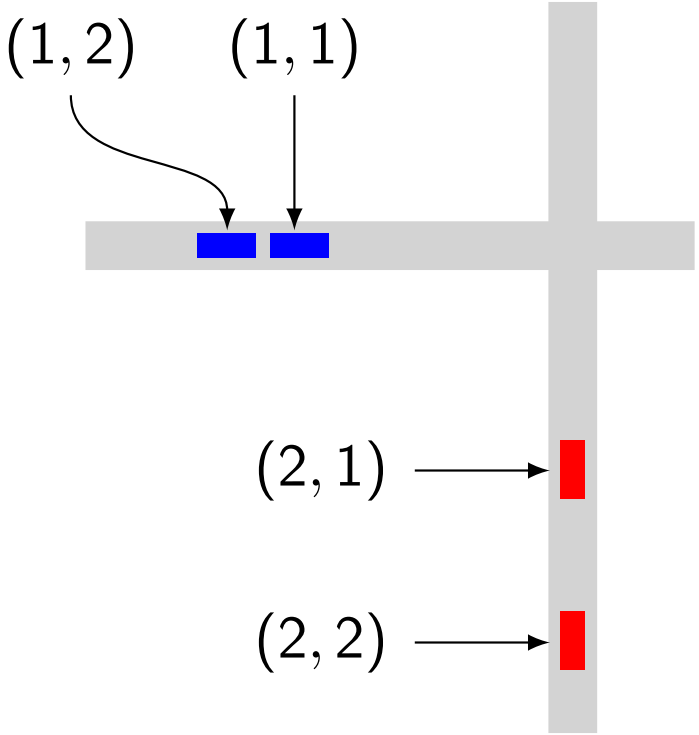
- Vehicle $i = (r, k)$ is the kth vehicle on route r



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- Vehicle $i = (r, k)$ is the kth vehicle on route r
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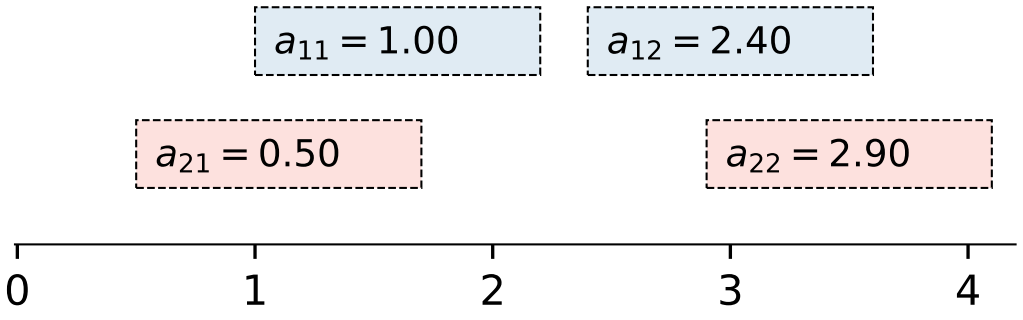

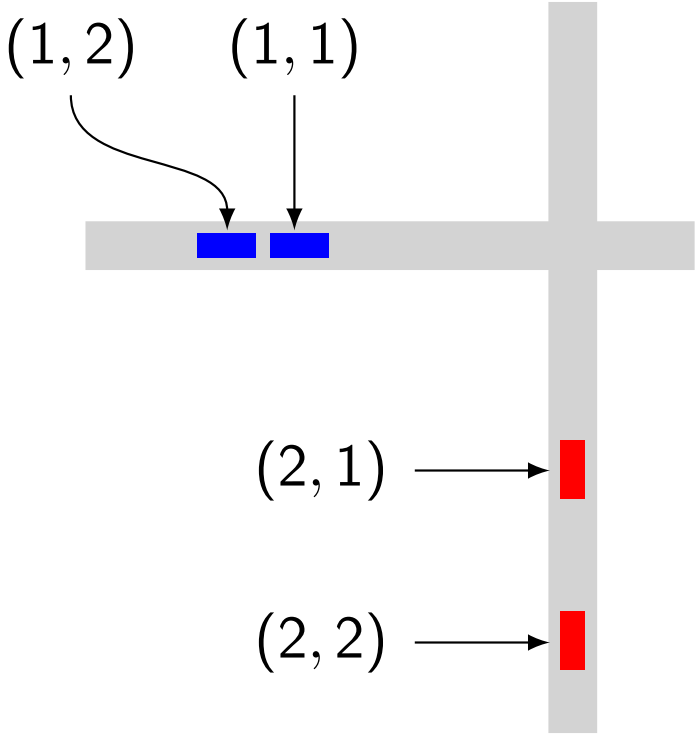

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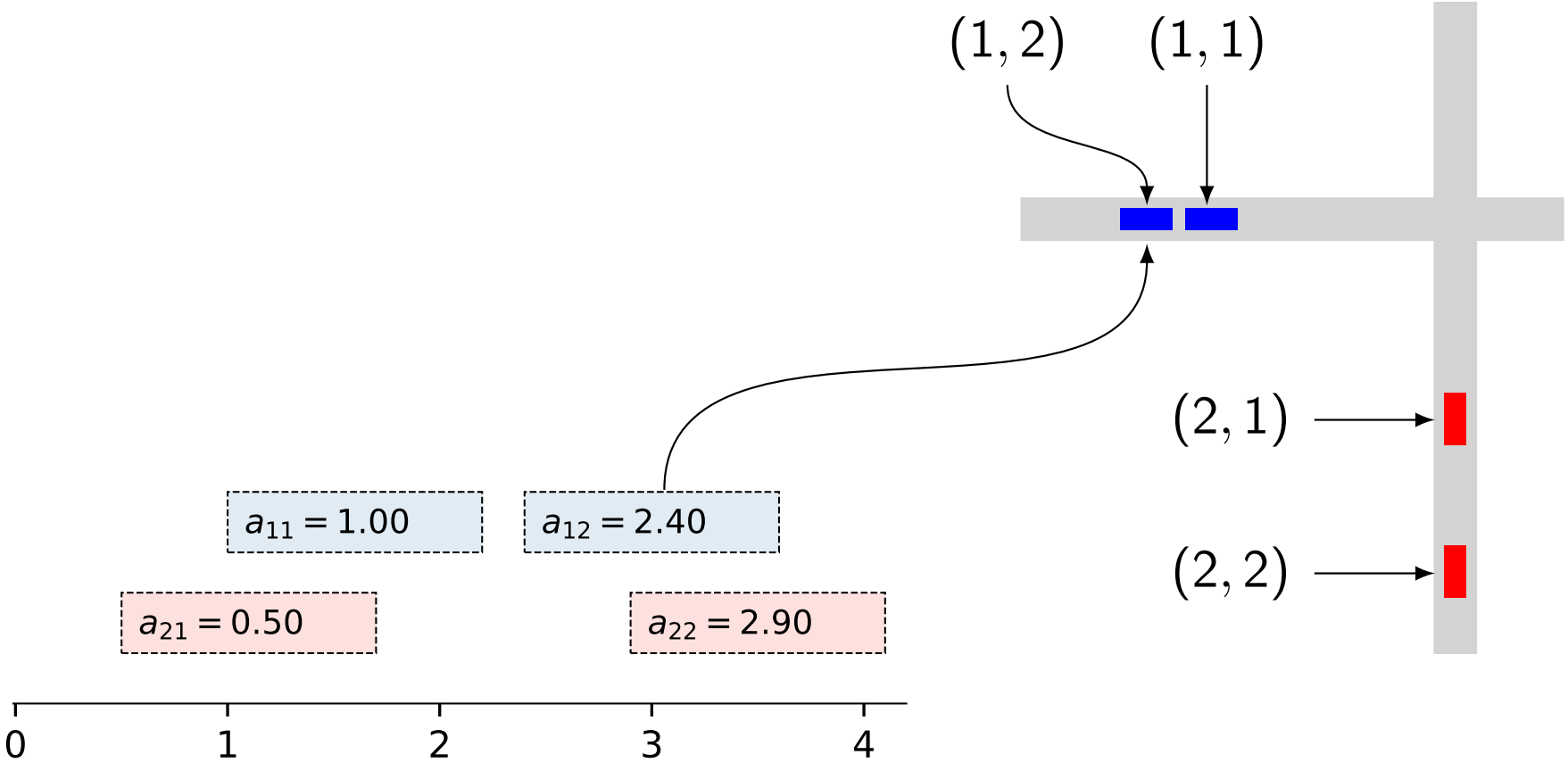

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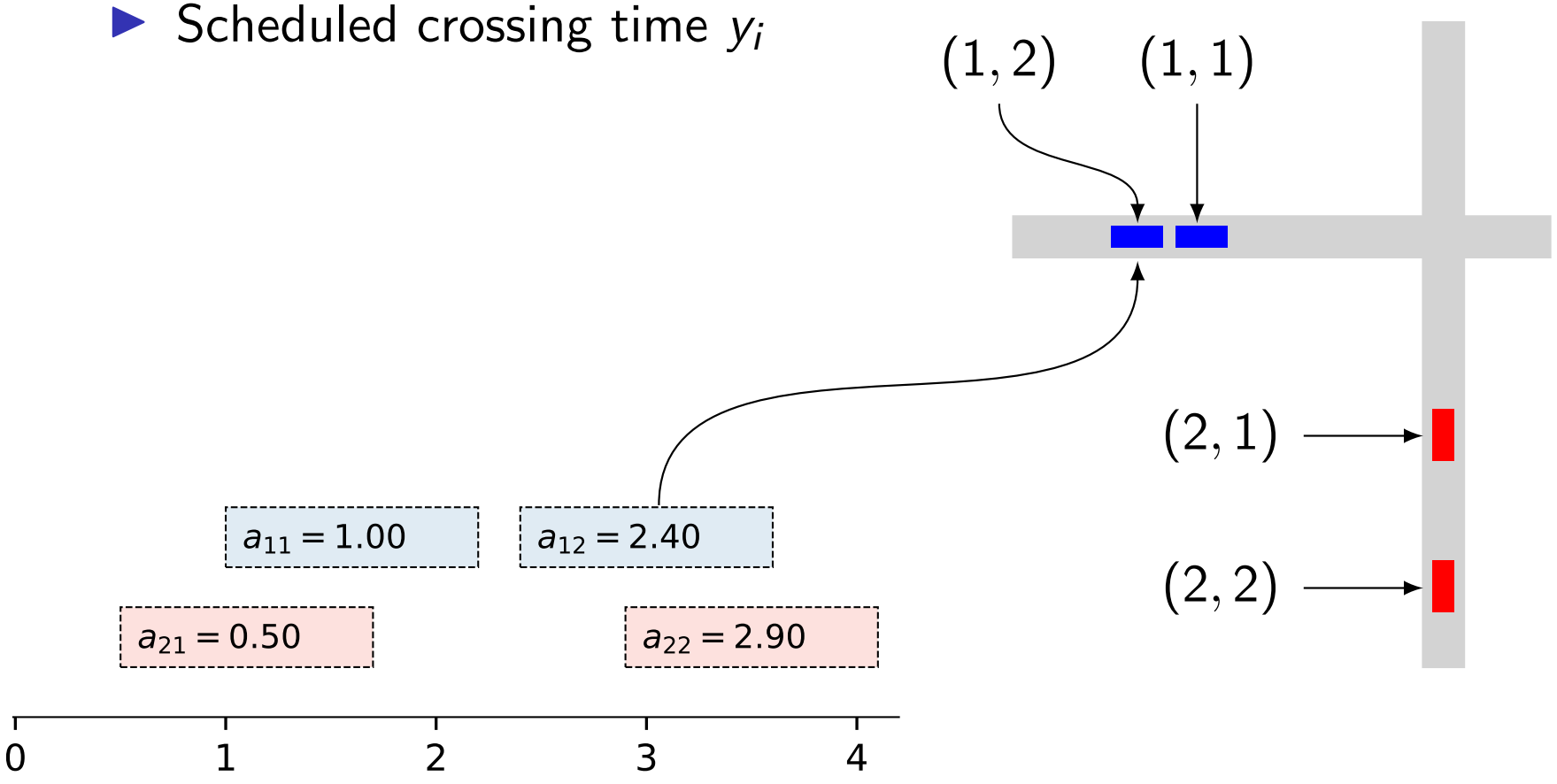

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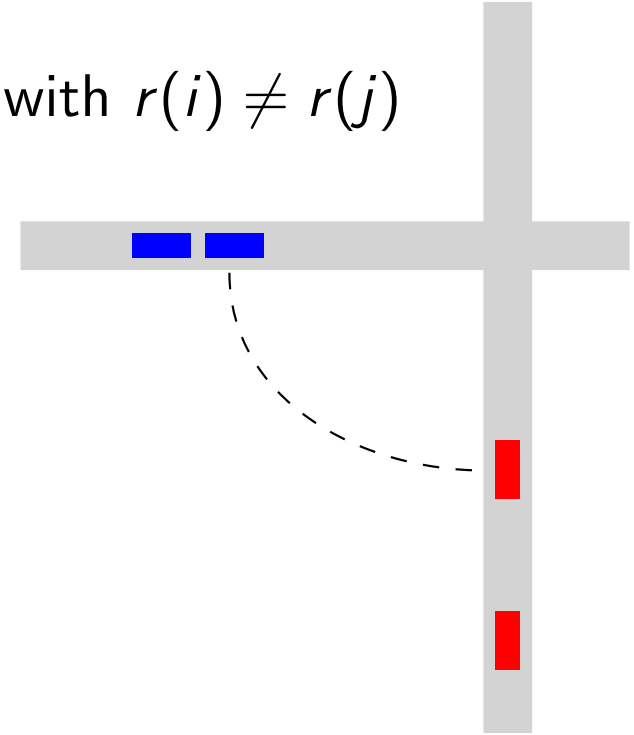

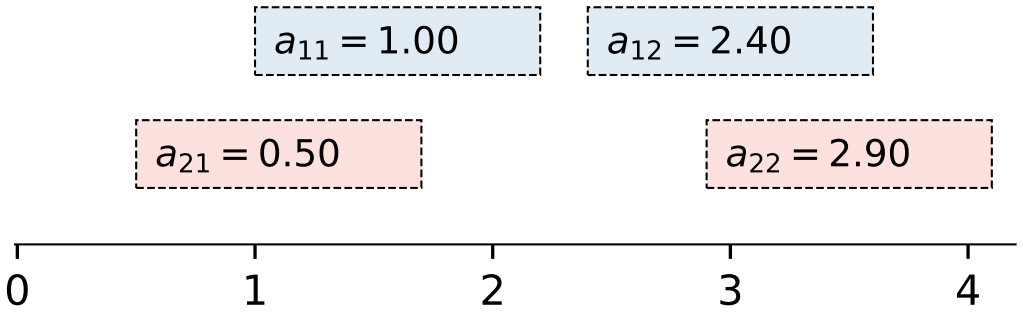

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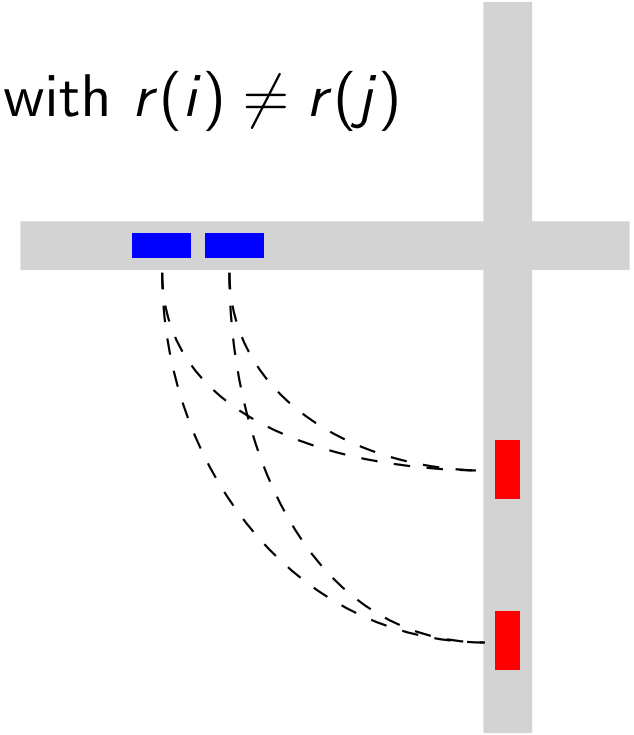

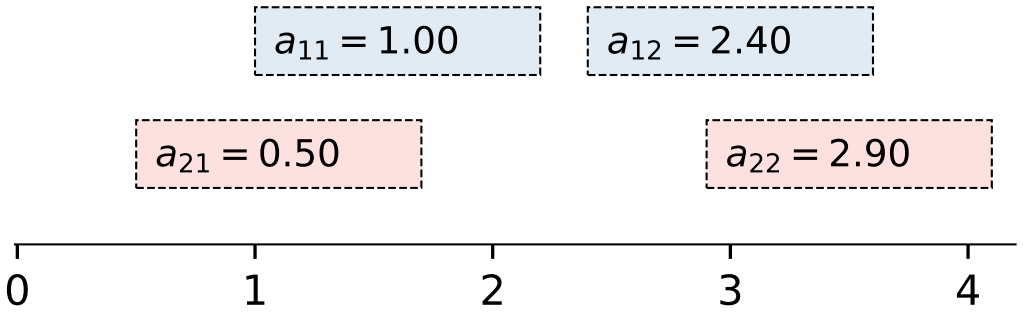

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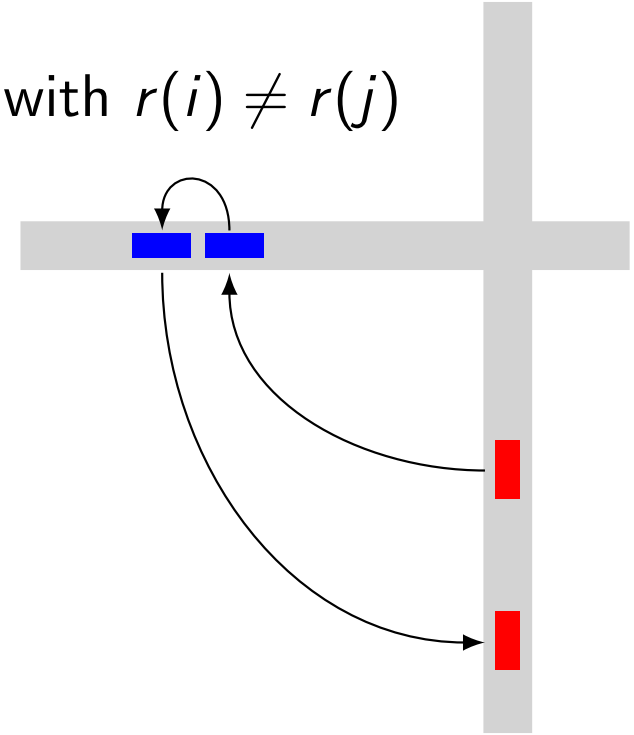

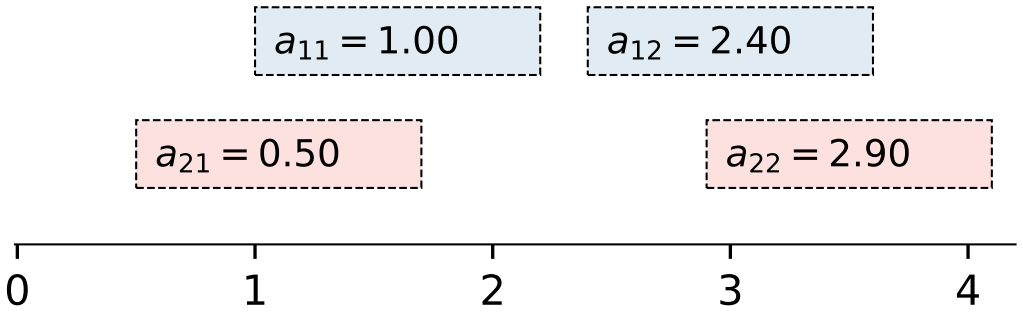

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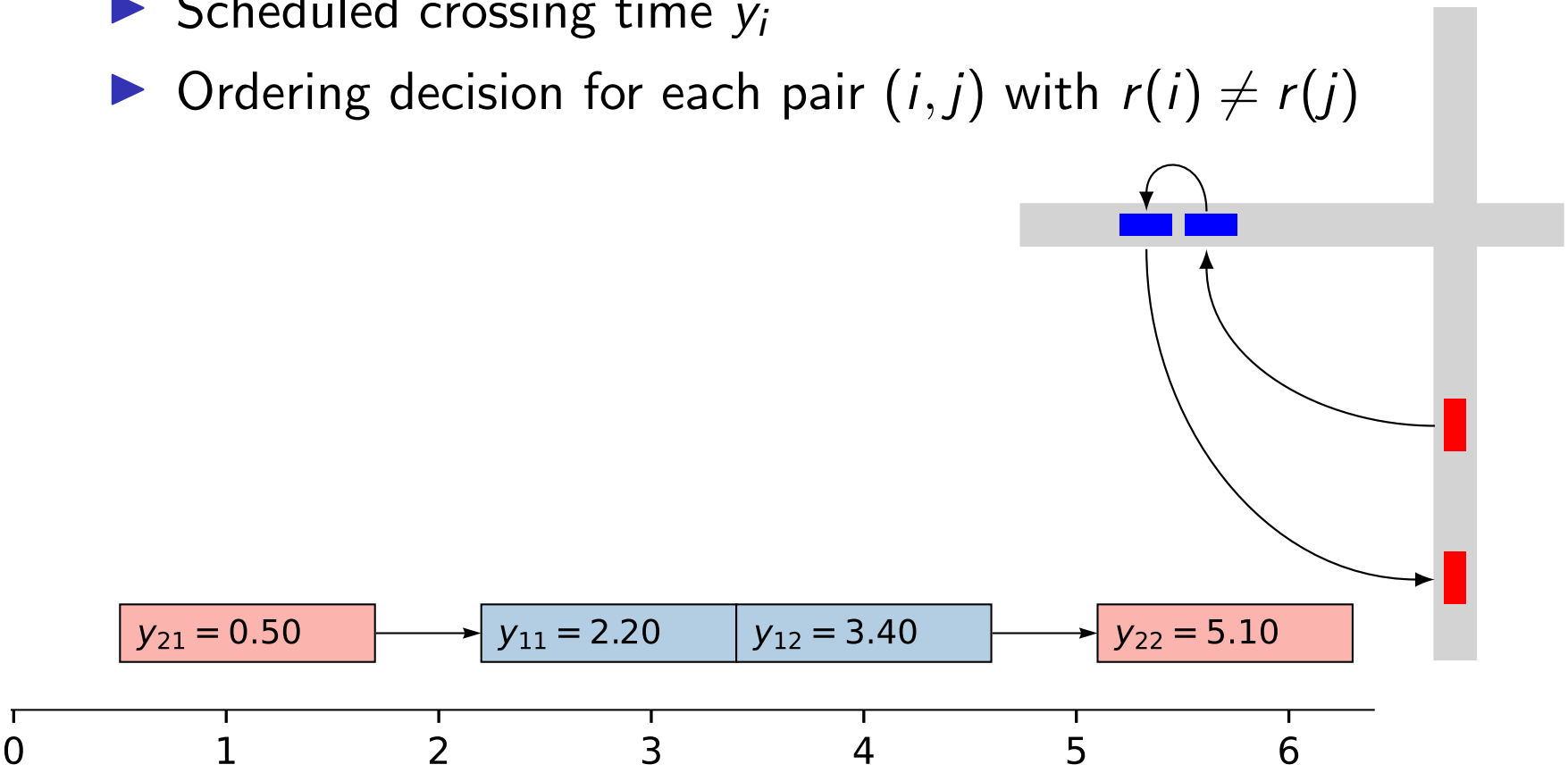

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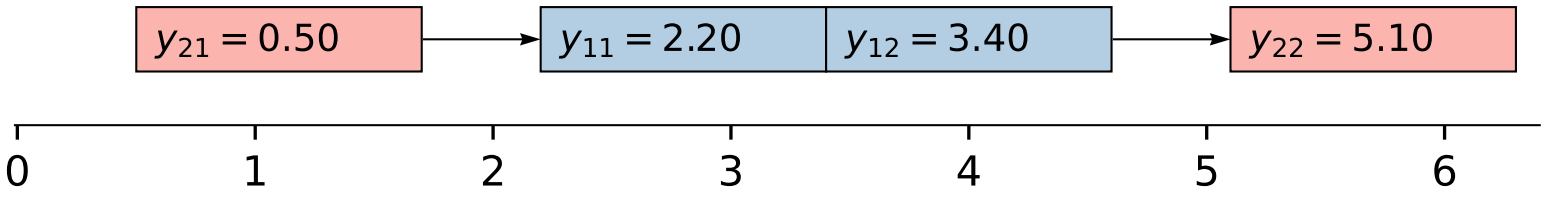

- Try to minimize total delay

$$\min_{y} \sum_{i} y_i - a_i$$

s.t. time slots are disjoint

Crossing Time Scheduling

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$$\text{s.t. time slots are disjoint}$$

- Ordering decisions $\rightarrow$ Binary variables
- Mixed-Integer Linear Program (MILP)

Crossing Time Scheduling

- How fast can we solve this?

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- A couple of options
  - Reformulation
  - Primal heuristics
  - Cutting planes
  - Better branching decisions

Crossing Time Scheduling

- How fast can we solve this?
- Vanilla integer programming does not scale so well

- A couple of options
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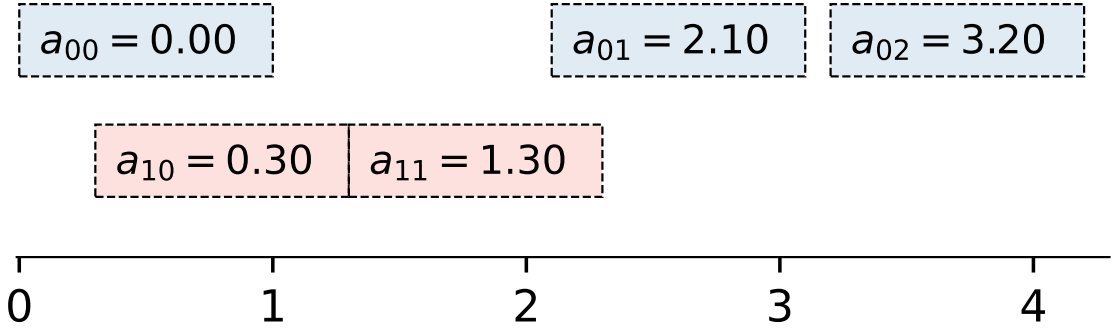
Cutting Planes in Single Intersection

Platoon Preservation



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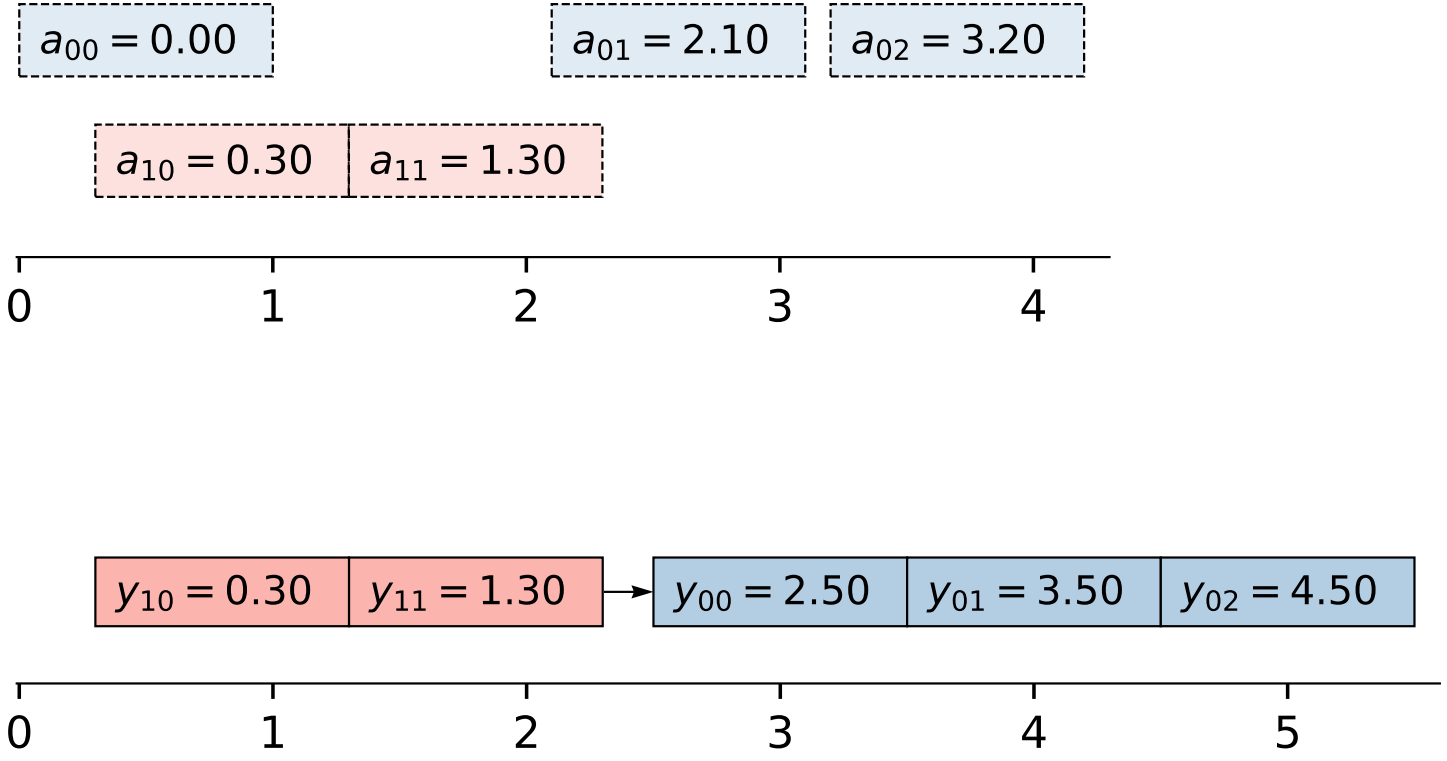

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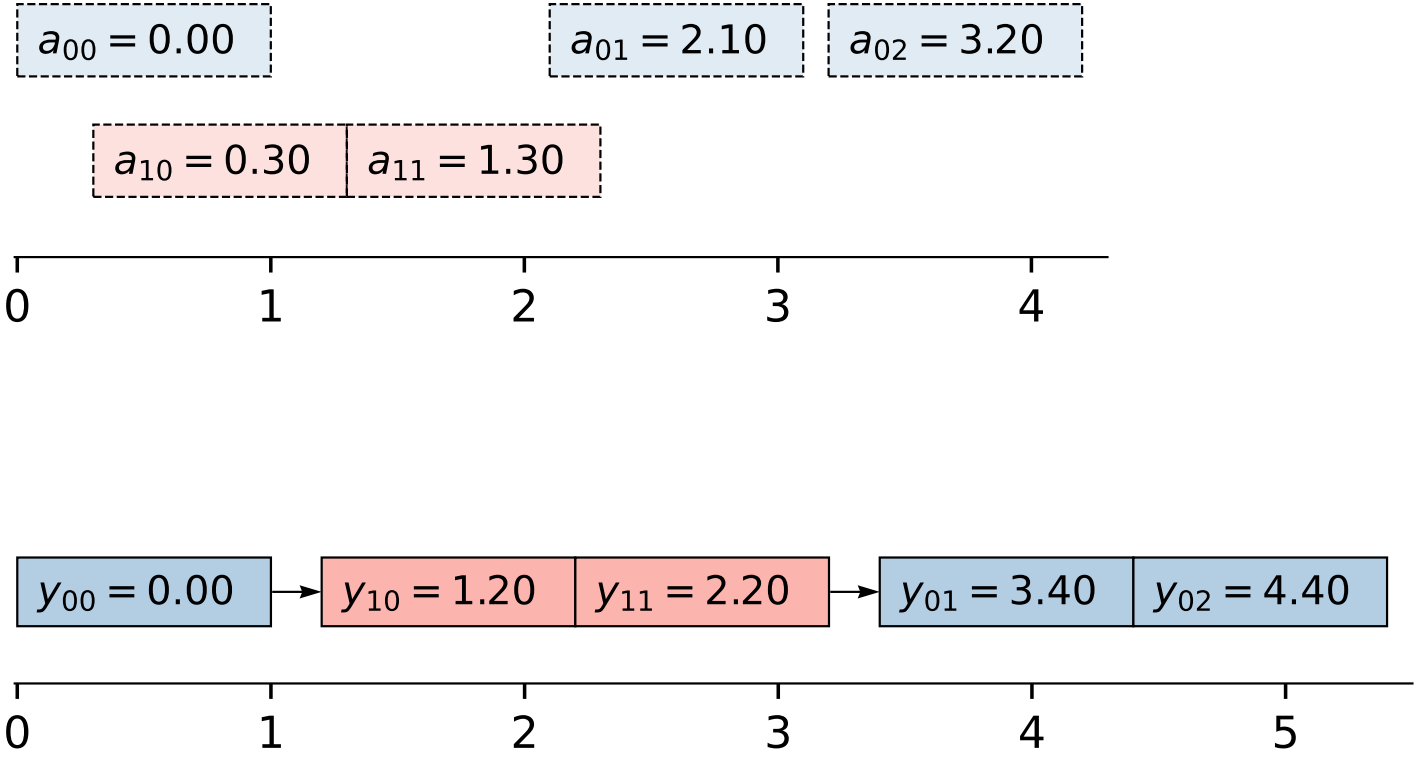

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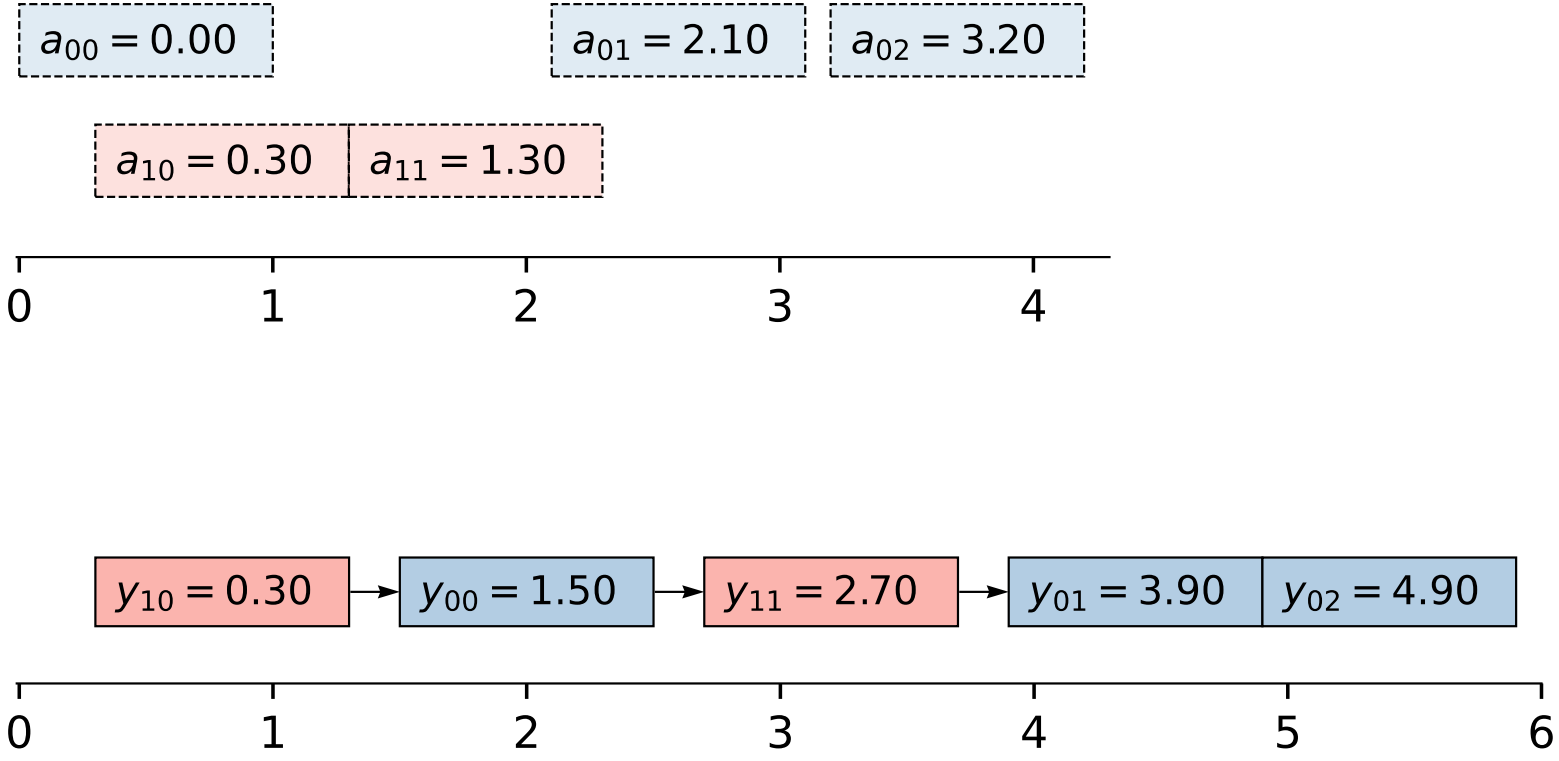

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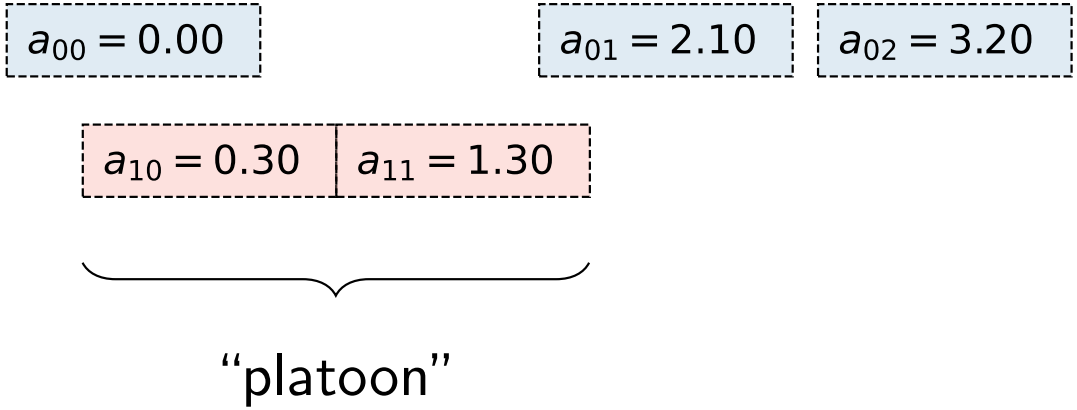
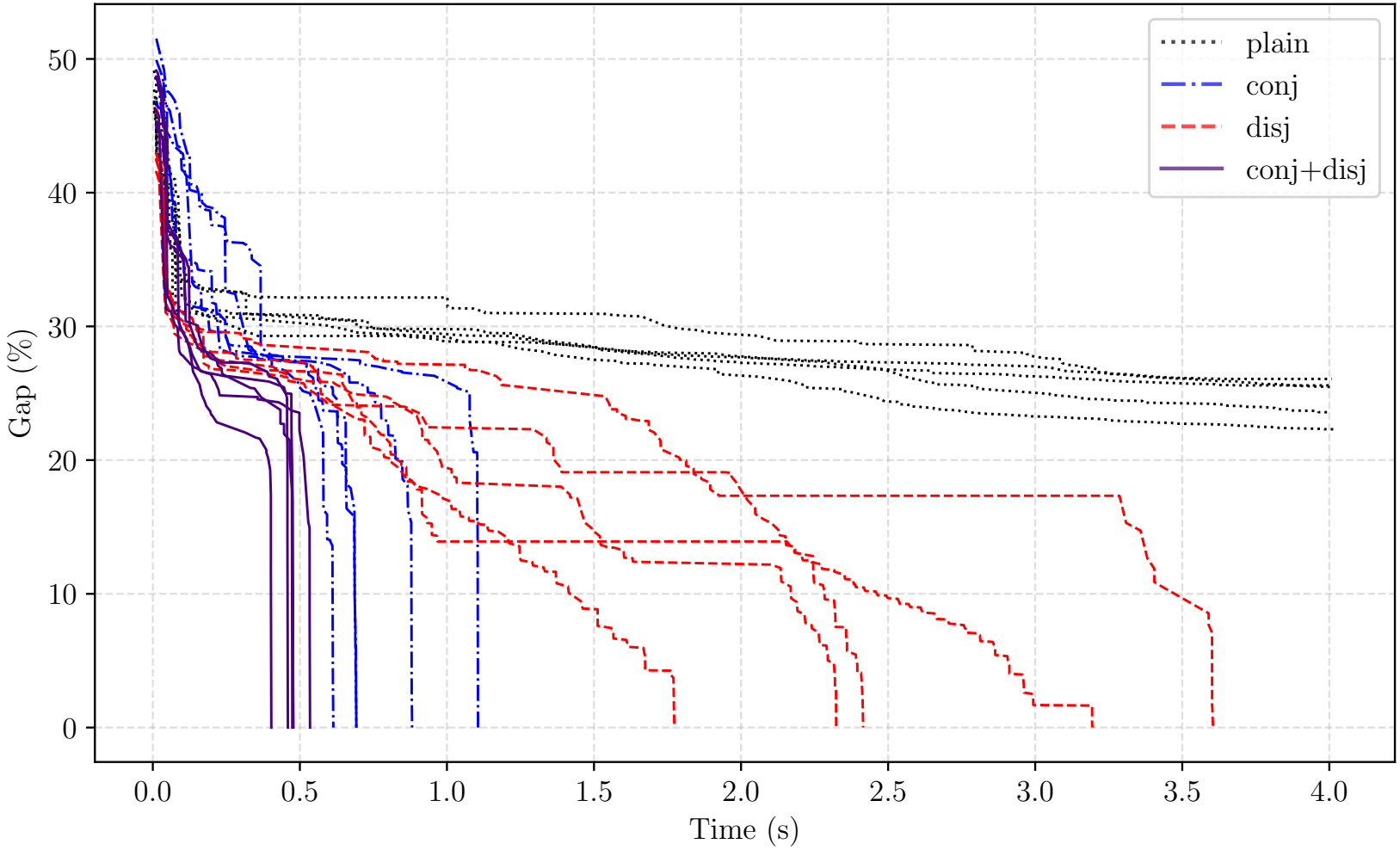


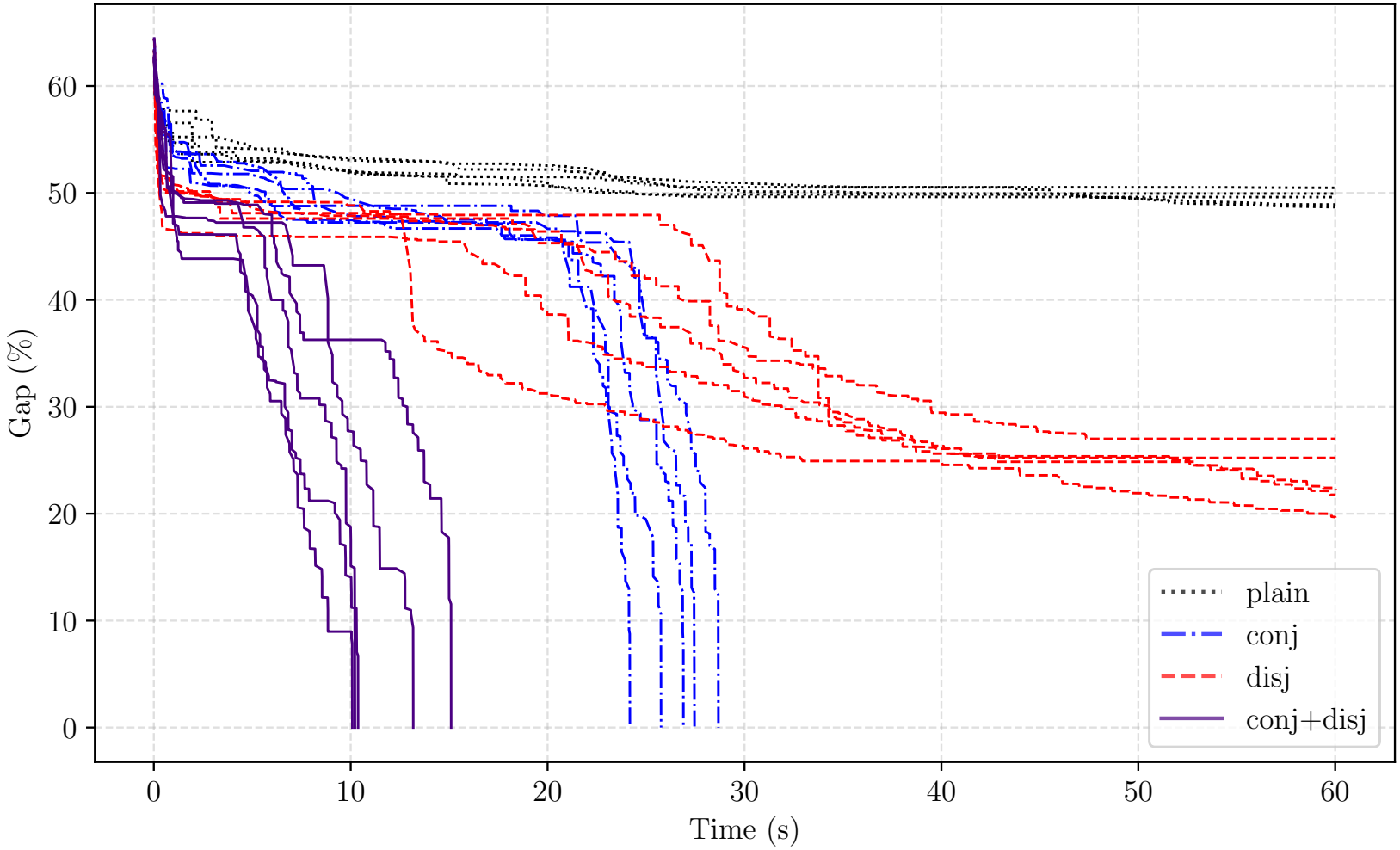
Cutting Planes in Single Intersection

- Two routes, 20 vehicles per route, 40 in total



Cutting Planes in Single Intersection

- Three routes, 20 vehicles per route, 60 in total

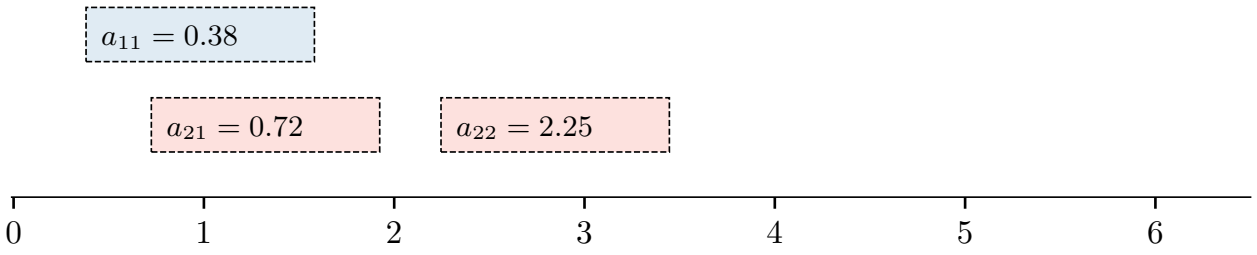


Heuristics and Combinatorial Learning

Constructive Heuristics

- Construct the schedule step by step

Constructive Heuristics



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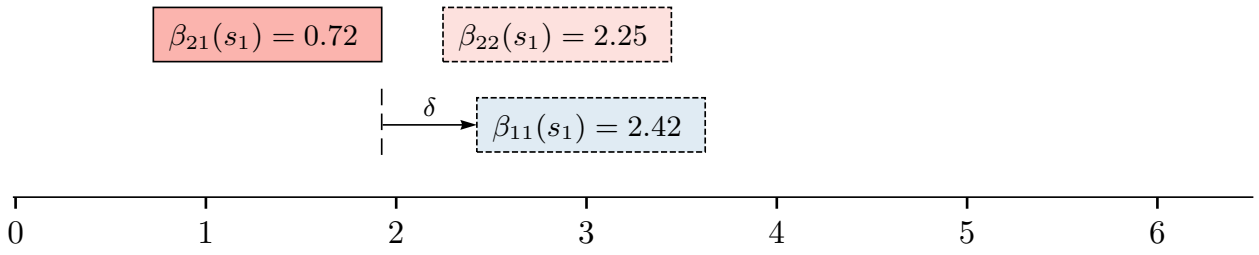

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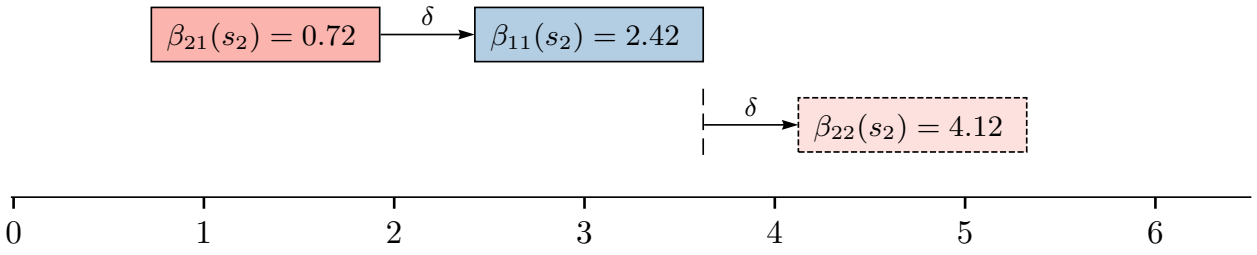

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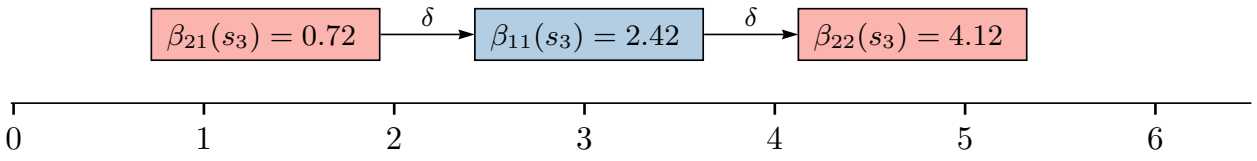

- Imitation learning
- Policy-gradient optimization

Takeaways

Key takeaway

Beamer helps create professional presentations.